

Pericyclic reactions 1: cycloadditions

34

Connections

➡ Building on

- Structure of molecules [ch4](#)
- Reaction mechanisms [ch5](#)
- Conjugation and delocalization [ch7](#)
- Reactions of alkenes [ch19](#) & [ch22](#)
- Aromatic heterocycles [ch29](#) & [ch30](#)

Arriving at

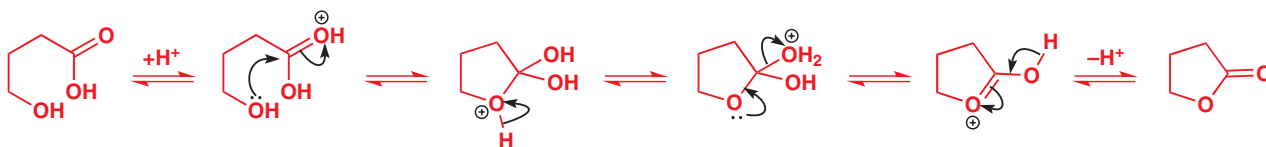
- In cycloadditions electrons move in a ring
- In cycloadditions more than one bond is formed simultaneously
- There are no intermediates in cycloadditions
- Cycloadditions are a type of pericyclic reaction
- The rules that govern cycloadditions: how to predict what will and will not work
- Photochemical reactions: reactions that need light
- Making six-membered rings by the Diels–Alder reaction
- Making four-membered rings by [2 + 2] cycloaddition
- Making five-membered rings by 1,3-dipolar cycloaddition
- Using cycloaddition to functionalize double bonds stereospecifically
- Using ozone to break C=C double bonds

➡ Looking forward to

- Electrocyclic reactions and sigmatropic rearrangements [ch35](#)
- Radical reactions [ch37](#)
- Reactions of carbenes [ch38](#)
- Asymmetric synthesis [ch41](#)

A new sort of reaction

Most organic reactions are ionic. Electrons move from an electron-rich atom towards an electron-poor atom: anions or cations are intermediates. Formation of a cyclic ester (a lactone) is an example. The reaction involves five steps and four intermediates. The reaction is acid-catalysed and each intermediate is a cation. Electrons flow in one direction in each step—towards the positive charge. This is an ionic reaction.



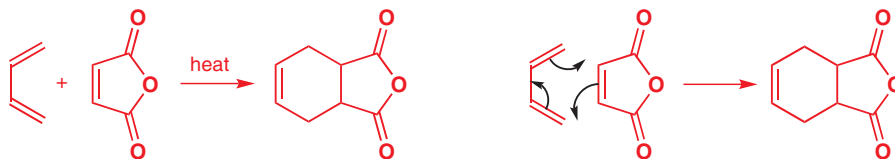
This chapter is about a totally different reaction type. Electrons move round a circle and there are no positive or negative charges on any intermediates—indeed, there are no inter-

■ In Chapter 24 you met a brief introduction to a third category—radical reactions—in which one electron instead of two is on the move. This will be developed in more detail in Chapter 37.

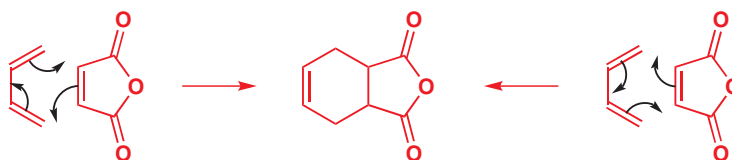
Otto Diels (1876–1954) and his research student Kurt Alder (1902–58) worked at the University of Kiel and discovered this reaction in 1928. They won the Nobel Prize in 1950. Diels also discovered carbon suboxide, C_2O_3 (see p. 420).

■ Cycloadditions are the first of three classes of pericyclic reactions, and the whole of this chapter will be devoted to cycloadditions. The other two—sigmatropic and electrocyclic reactions—are discussed in Chapter 35.

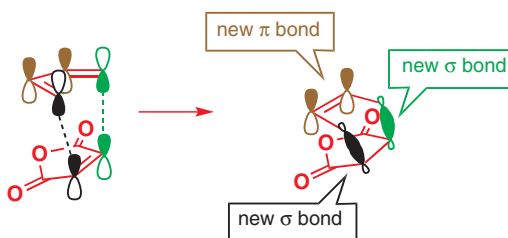
mediates at all. This type of reaction is called **pericyclic**. The most famous example is the **Diels–Alder reaction**. This reaction goes in a single step simply on heating. We can draw the mechanism with the electrons going round a six-membered ring.



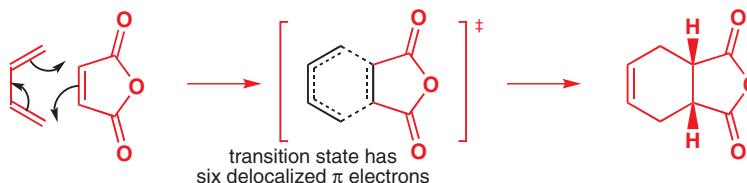
Each arrow leads directly to the next, and the last arrow connects to the first. We have drawn the electrons rotating clockwise, but it would make no difference at all if we drew the electrons rotating anticlockwise.



Both mechanisms are equally correct. The electrons do not really rotate at all. In reality two π bonds disappear and two σ bonds take their place by the electrons moving smoothly out of the π orbitals into the σ orbitals. Such a reaction is called a **cycloaddition**. We must spend some time working out how this could happen. First, just consider the orbitals that overlap to form the new bonds. Providing the reagents approach in the right way, nothing could be simpler.



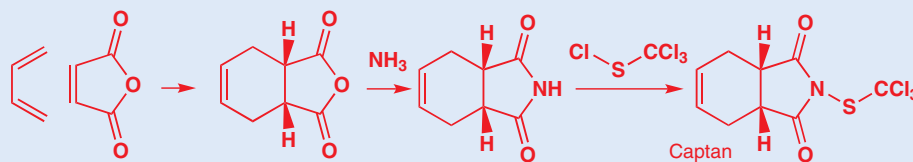
The black p orbitals are perfectly aligned to make a new σ bond, as are the two green orbitals. As this is a one-step reaction there are no intermediates but there is one transition state looking something like this:



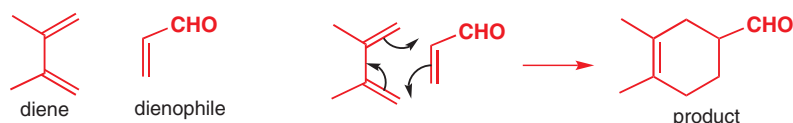
One reason that the Diels–Alder reaction goes so well is that the transition state has six delocalized π electrons and thus is aromatic in character, having some of the special stabilization of benzene. You could look at it as a benzene ring having all its π bonds but missing two σ bonds. This simple picture is fine as far as it goes, but it is incomplete. We shall return to a more detailed orbital analysis when we have described the reaction in more detail.

Captan

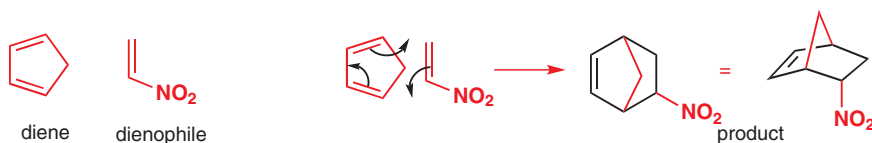
One important industrial application of the Diels–Alder reaction we have been discussing is in the synthesis of the agricultural fungicide Captan.

**General description of the Diels–Alder reaction**

Diels–Alder reactions occur between a **conjugated diene** and an alkene, usually called the **dienophile**. Here are some examples: first an open-chain diene with a simple unsaturated aldehyde as the dienophile.

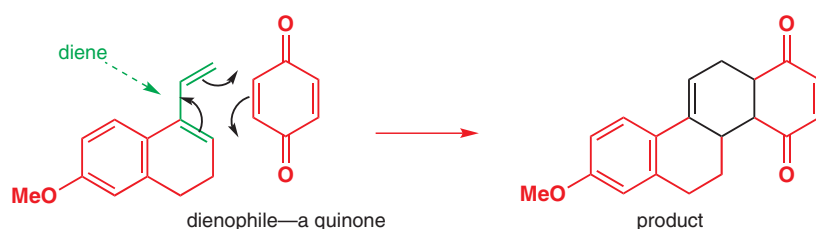


The mechanism is the same and a new six-membered ring is formed having one double bond. Now a reaction between a cyclic diene and a nitroalkene.



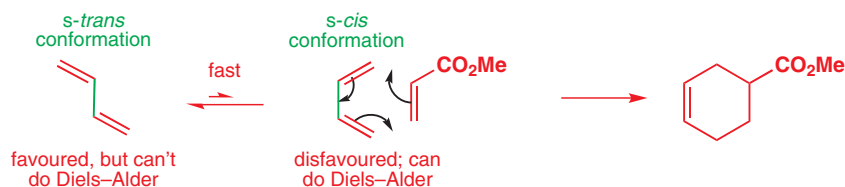
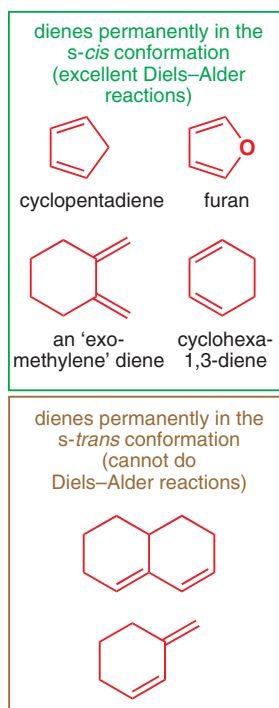
➡ This fused ring system, and how to draw it, was described in more detail in Chapter 32.

The mechanism leads clearly to the first drawing of the product but this is a cage structure and the second drawing is better. The new six-membered ring is outlined in black in both diagrams. A more elaborate example shows that quite complex molecules can be quickly assembled with this wonderful reaction.

**The diene**

The diene component in the Diels–Alder reaction can be open-chain or cyclic and it can have many different kinds of substituents. There is only one limitation: it must be able to take up the conformation shown in the mechanism. Butadiene normally prefers the *s-trans* conformation with the two double bonds as far away from each other as possible for steric reasons. The barrier to rotation about the central σ bond is small (about 30 kJ mol⁻¹ at room temperature) and rotation to the less favourable but reactive *s-cis* conformation is rapid.

■ The 's' in the terms '*s-cis*' and '*s-trans*' refers to a σ bond and indicates that these are conformations about a single bond and not configurations about a double bond.



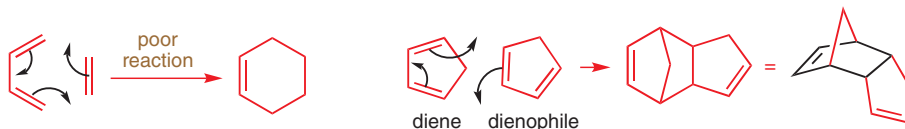
Cyclic dienes that are permanently in the *s-cis* conformation are exceptionally good at Diels–Alder reactions—cyclopentadiene is a classic example—but cyclic dienes that are permanently in the *s-trans* conformation and cannot adopt the *s-cis* conformation will not do the Diels–Alder reaction at all. The two ends of these dienes cannot get close enough to react with an alkene and, in any case, the product would have an impossible *trans* double bond in the new six-membered ring. (In the Diels–Alder reaction, the old σ bond in the centre of the diene becomes a π bond in the product and the conformation of that σ bond becomes the configuration of the new π bond in the product.)

• The diene

The diene must have the *s-cis* conformation.

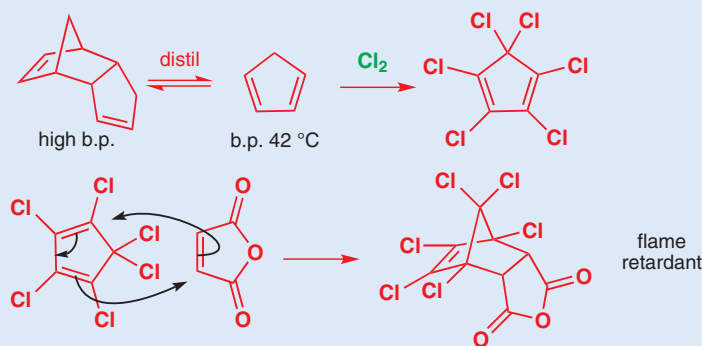
The dienophile

The dienophiles you have seen in action so far all have one thing in common. They have an electron-withdrawing group conjugated to the alkene. This is a common, although not exclusive, feature of Diels–Alder dienophiles. There must be some extra conjugation—at least a phenyl group or a chlorine atom—or the cycloaddition does not occur. You will often see the reaction between butadiene and a simple alkene (even ethylene) given in books as the basic Diels–Alder reaction. This occurs in only poor yield. Attempts to combine even such a reactive diene as cyclopentadiene with a simple alkene lead instead to the dimerization of the diene. One molecule acts as the diene and the other as the dienophile to give the cage structure shown.



Cyclopentadiene

Cyclopentadiene is formed in considerable amounts during the refining of petroleum. It exists as its dimer at room temperature but can be dissociated into the monomer on heating—the effect of the increased importance of entropy at higher temperatures (Chapter 12). It can be chlorinated to give hexachlorocyclopentadiene, and the Diels–Alder product of this diene with maleic anhydride is a flame retardant.

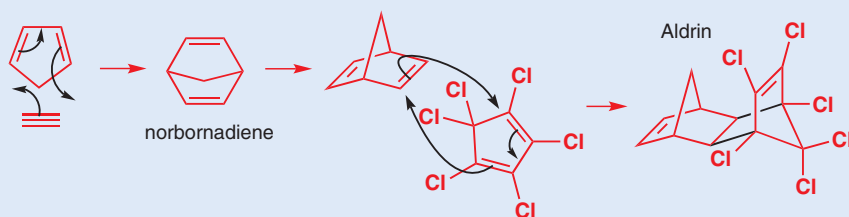


Simple alkenes that do undergo the Diels–Alder reaction include conjugated carbonyl compounds, nitro compounds, nitriles, sulfones, aryl alkenes, vinyl ethers and esters, haloalk-

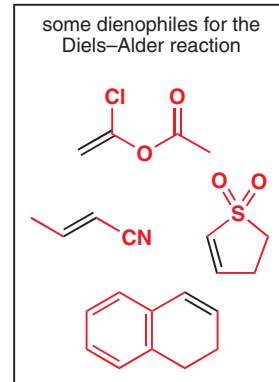
enes, and dienes. In addition to those you have seen so far, a few examples are shown in the margin. In the last example it is the isolated double bond in the right-hand ring that accepts the diene. Conjugation with the left-hand ring activates this alkene. But what exactly do we mean by ‘activate’ in this sense? We shall return to that question in a minute.

Dieldrin and Aldrin

In the 1950s two very effective pesticides were launched and their names were ‘Dieldrin’ and ‘Aldrin’. As you may guess they were made by the Diels–Alder reaction. Aldrin is derived from two consecutive Diels–Alder reactions. In the first, cyclopentadiene reacts with acetylene to give a simple symmetrical cage molecule ‘norbornadiene’ (bicyclo[2.2.1]hepta-2,5-diene). Norbornadiene is not conjugated and cannot take part in a Diels–Alder reaction as a *diene*. However, it is quite strained because of the cage and it reacts as a *dienophile* with perchlorocyclopentadiene to give Aldrin.



This is quite a complex product but we hope you can see how it is made up by looking at the two new bonds marked in black. Dieldrin is the epoxide of Aldrin. The use of these compounds, like that of many organochlorine compounds, was eventually banned when it was found that chlorine residues were accumulating in the fat of animals high up in the food chain, such as birds of prey and humans.

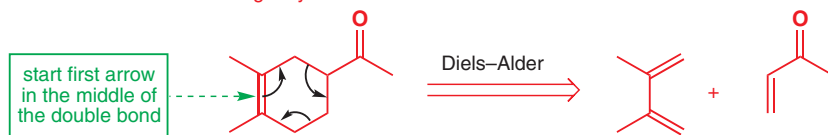


The product

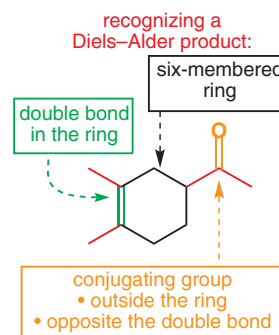
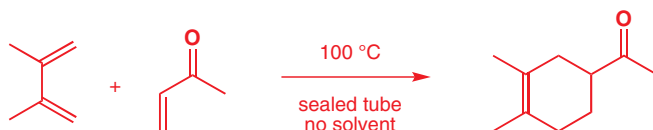
Recognizing a Diels–Alder product is straightforward. Look for the six-membered ring, the double bond inside the ring, and the conjugating group outside the ring and on the opposite side of the ring from the alkene. These three features mean that the compound is a possible Diels–Alder product.

The simplest way to find the starting materials is to carry out a disconnection that is closer to a real reaction than most. Just draw the reverse Diels–Alder reaction. To do this, draw three arrows going round the cyclohexene ring, starting the first arrow in the middle of the double bond. It doesn't, of course, matter which way round you go.

the disconnection is the imaginary reverse Diels–Alder reaction



The reaction couldn't be simpler—just heat the components together without solvent or catalyst. Temperatures of around 100–150°C are often needed and this may mean using a sealed tube if the reagents are volatile, as here.

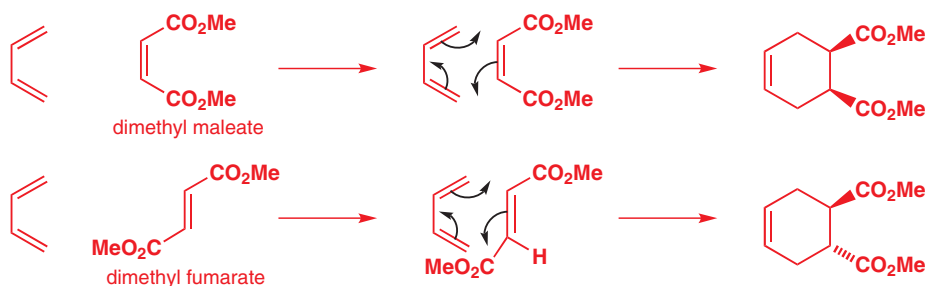


Disconnections and retrosynthetic arrows of the type shown here are ways of thinking about how to make molecules. They appear throughout Chapter 28.

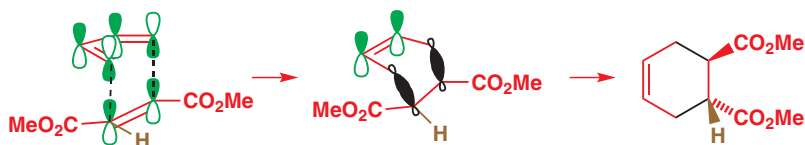
Stereochemistry

The Diels–Alder reaction is stereospecific. If there is stereochemistry in the dienophile, then it is faithfully reproduced in the product. Thus *cis* and *trans* dienophiles give different diastereoisomers of the product. Esters of maleic and fumaric acids provide a simple example.

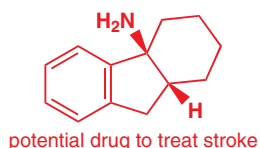
Interactive explanation of the effect of dienophile stereochemistry



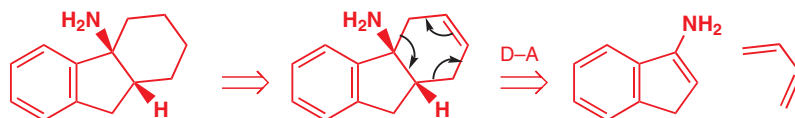
In both cases the ester groups simply stay where they are. They are *cis* in the dienophile in the first reaction and remain *cis* in the product. They are *trans* in the dienophile in the second reaction and remain *trans* in the product. The second example may look less convincing—may we remind you that the diene actually comes down on top of the dienophile like this:



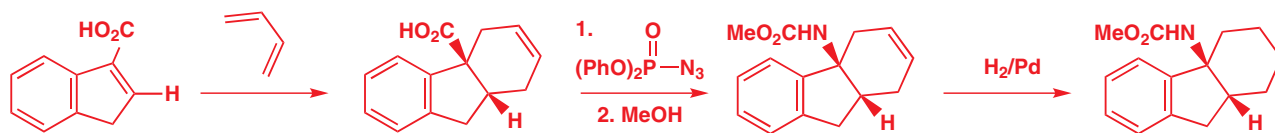
One of the CO_2Me groups is tucked under the diene in the transition state and then, when the product molecule is flattened out in the last drawing, that CO_2Me group appears underneath the ring. The brown hydrogen atom remains *cis* to the other CO_2Me group.



The search by the Parke–Davis company for drugs to treat strokes provided an interesting application of dienophile stereochemistry. The kinds of compound they wanted were tricyclic amines. They don't look like Diels–Alder products at all. But if we insert a double bond in the right place in the six-membered ring, Diels–Alder (D–A) disconnection becomes possible.



Butadiene is a good diene, but the enamine required is not a good dienophile. An electron-withdrawing group such as a carbonyl or nitro group is preferable: either would do the job. In the event a carboxylic acid that could be converted into the amine by a rearrangement with $(\text{PhO})_2\text{PON}_3$ was used.



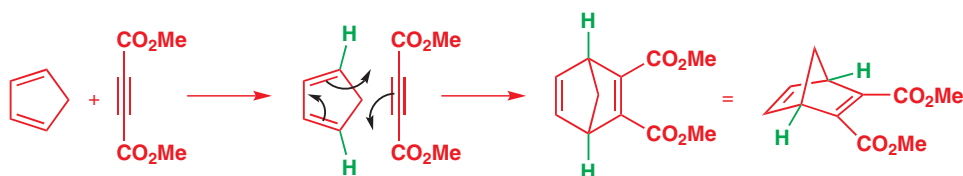
■ The rearrangement with $(\text{PhO})_2\text{PON}_3$ is a Curtius rearrangement: it is described in Chapter 38.

■ You can add the Diels–Alder reaction to your mental list of reactions to consider for making a single diastereoisomer from a single geometrical isomer of an alkene: see Chapter 33.

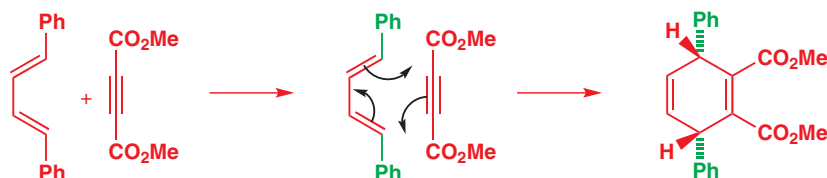
The stereochemistry at the ring junction must be *cis* because the cyclic dienophile can have only a *cis* double bond. Hydrogenation removes the double bond in the product and shows just how useful the Diels–Alder reaction is for making saturated rings, particularly when there is some stereochemistry to be controlled.

Stereochemistry of the diene

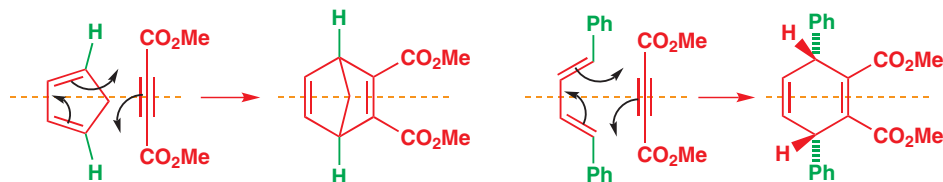
This is slightly more complicated as the diene can be *cis,cis*, or *cis,trans* (there are two of these if the diene is unsymmetrical), or *trans,trans*. We shall look at each case with the same dienophile, an acetylene dicarboxylate, as there is then no stereochemistry in the triple bond! Starting with *cis,cis*-dienes is easy if we make the diene cyclic.



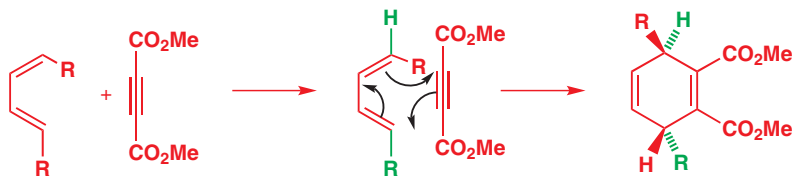
The diene has two sets of substituents—inside and outside. The inside one is the bridging CH_2 group and it has to end up on one side of the molecule (above in the last diagram) while the two green hydrogens are outside and remain so. In the final diagram they are below the new six-membered ring. With a *trans,trans*-diene we simply exchange the two sets of substituents, in this example putting Ph where H was and putting H where the bridging CH_2 group was. This is the reaction:



The green Ph groups end up where the hydrogens were in the first example—beneath the new six-membered ring—and the hydrogens end up above. It may seem puzzling at first that a *trans,trans*-diene gives a product with the two phenyls *cis*. Another way to look at these two reactions is to consider their symmetry. Both have a plane of symmetry throughout and the products must have this symmetry too because the reaction is concerted and no significant movement of substituents can occur. The orange dotted line shows the plane of symmetry, which is at right angles to the paper.



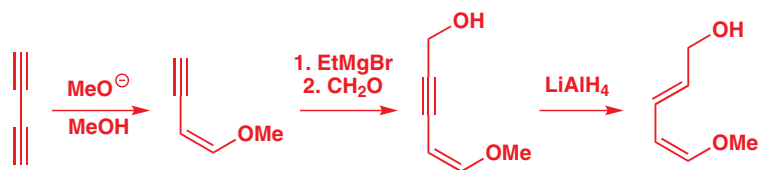
The remaining case—the *cis,trans*-diene—is rarer than the first two, but is met sometimes. This unsymmetrical diene means the two substituents clearly end up on opposite sides of the new six-membered ring.



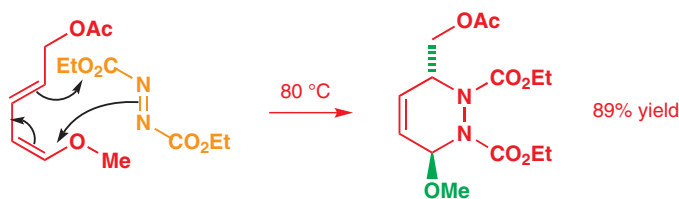
The red R group may seem to get in the way of the reaction but, of course, the dienophile is not approaching in the plane of the diene but from underneath. It is difficult to find a convincing example of this stereochemistry as there are so few known, partly because of the difficulty of making *E,Z* dienes. One good approach uses two reactions you met in Chapter 29 for the control of double-bond geometry. The *cis* double bond is put in first by the addition of methanol to butadiyne and the *trans* double bond then comes from LiAlH_4 reduction of the intermediate acetylenic alcohol.

➡ The mechanism for these reactions is given on pp. 682 and 684.

➡ DEAD is a key component of the Mitsunobu reaction: see p. 349.



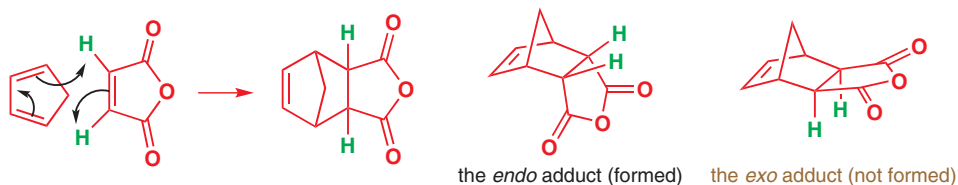
The acetate of this alcohol is used in a Diels–Alder reaction with the interesting dienophile DEAD (diethyl azodicarboxylate—in orange). The product is formed in excellent yield and has the *trans* stereochemistry that was predicted. The amide nitrogen atoms are planar, so there is no question of stereochemistry there. DEAD itself can equilibrate rapidly between *E* and *Z* isomers, but the *E* predominates.



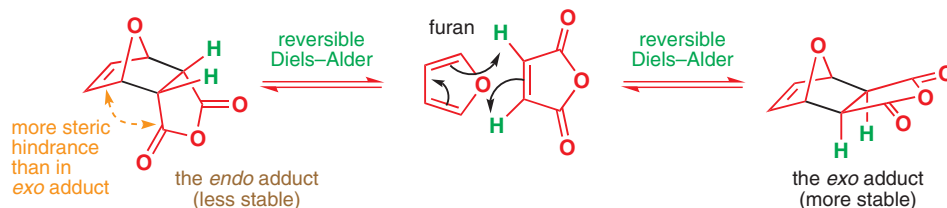
Now to the most interesting cases of all, when both the diene and the dienophile have stereochemistry.

The *endo* rule for the Diels–Alder reaction

It is probably easier to see this when both the diene and the dienophile are cyclic. All the double bonds are *cis* and the stereochemistry is clearer. In the most famous Diels–Alder reaction of all time, that between cyclopentadiene and maleic anhydride, there are two possible products that obey all the rules we have so far described. They are the only possible diastereoisomers of the product—although it has four stereogenic centres, any other diastereoisomers would be impossibly strained.



The two green hydrogen atoms must be *cis* in the product, but now there are two such compounds, known as the *exo* and *endo* products. When the reaction is carried out, the product is, in fact, the *endo* compound. Only one diastereoisomer is formed, and it is the less stable one. How do we know this? Well, for cases in which the Diels–Alder reaction is reversible and therefore under thermodynamic control, the *exo* product is formed instead. The best known example results from the replacement of cyclopentadiene with furan in reaction with the same dienophile.



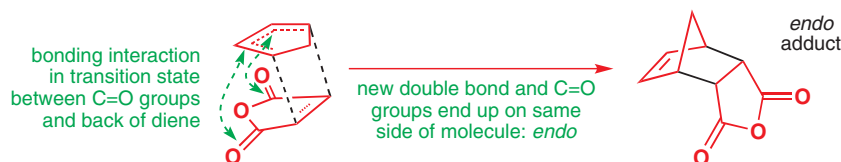
Why is the *exo* product the more stable? Look again at these two structures. On the left-hand side of the molecules, there are two bridges across the ends of the new bonds (highlighted in black): a one-C-atom bridge and a two-C-atom bridge. There is less steric hindrance if the smaller (that is, the one-atom) bridge eclipses the anhydride ring.

The *endo* product is less stable than the *exo* product and yet it is preferred in irreversible Diels–Alder reactions—it must be the kinetic product of the reaction. It forms faster because

■ These names arise from the relationship in space between the carbonyl groups on the dienophile and the newly formed double bond in the middle of the old diene. If these are on the same side they are called *endo* (inside) and if they are on opposite sides they are called *exo* (outside).

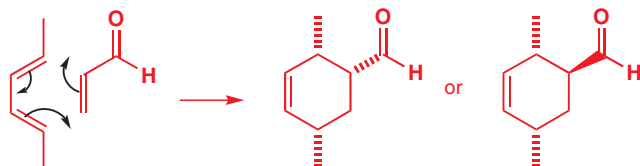
➡ Kinetic and thermodynamic control are discussed in Chapter 12, and you met Diels–Alder reactions with heterocycles in Chapter 30, p. 739.

a bonding interaction between the electron-deficient carbonyl groups of the dienophile and the developing π bond at the back of the diene lowers the energy of the transition state, leading to the *endo* product.



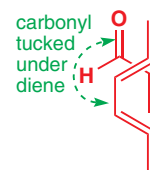
Interactive explanation of *endo* selectivity

The same result is found with acyclic dienes and dienophiles. Normally one diastereoisomer is preferred—the one with the carbonyl groups of the dienophile closest to the developing π bond at the back of the diene. Here is an example.

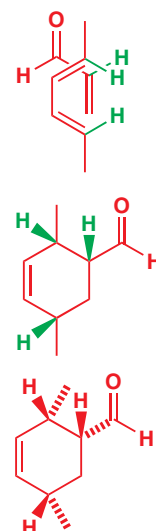


From our previous discussion (it's a *trans,trans* diene) we expect the two methyl groups to be *cis* to each other and the only question remaining is the stereochemistry of the aldehyde group—up or down? The aldehyde will be *endo*—but which compound is that? The easiest way to find the answer is to draw the reagents coming together in three dimensions. Here is one way to do this.

1. Draw the mechanism of the reaction and diagrams of the product to show what you are trying to decide. Put in the known stereochemistry if you wish. This we have just done (see above).
2. Draw both molecules in the plane of the paper with the diene on top and the carbonyl group of the dienophile tucked under the diene so it can be close to the developing π -bond.



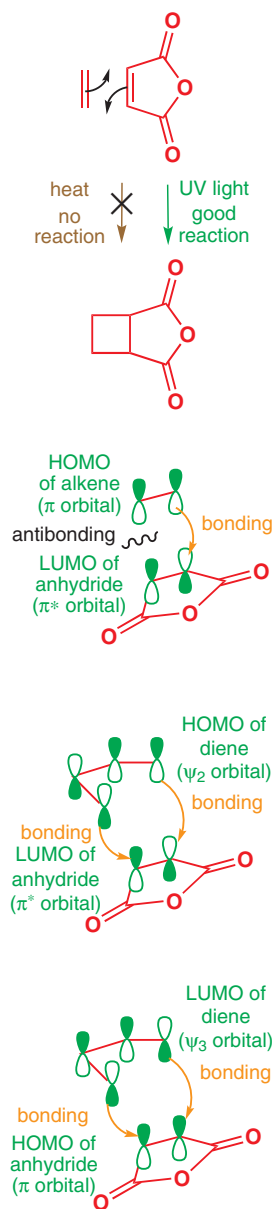
3. Now draw in all the hydrogen atoms on the carbon atoms that are going to become stereogenic centres, that is, those shown in green here.
4. Draw a diagram of the product. Unfold the molecule to show the six-membered ring. All the substituents to the right in the previous diagram are on one side of the new molecule. That is, all the green hydrogen atoms are *cis* to each other.
5. Draw a final diagram of the product with the stereochemistry of the other substituents shown too in the usual way. This is the *endo* product of the Diels–Alder reaction.



Time for some explanations

We have accumulated rather a lot of unexplained results.

- Why does the Diels–Alder reaction work so well?
- Why must we have a conjugating group on the dienophile?
- Why is the stereochemistry of each component retained so faithfully?
- Why is the *endo* product preferred kinetically?



■ You may need to remind yourself about the orbitals of conjugated π systems by re-reading Chapter 7.

There is more. The simpler picture we met earlier in this chapter also fails to explain why the Diels–Alder reaction occurs simply on heating while attempted additions of simple alkenes (rather than dienes) to maleic anhydride fail on heating but succeed under irradiation with UV light.

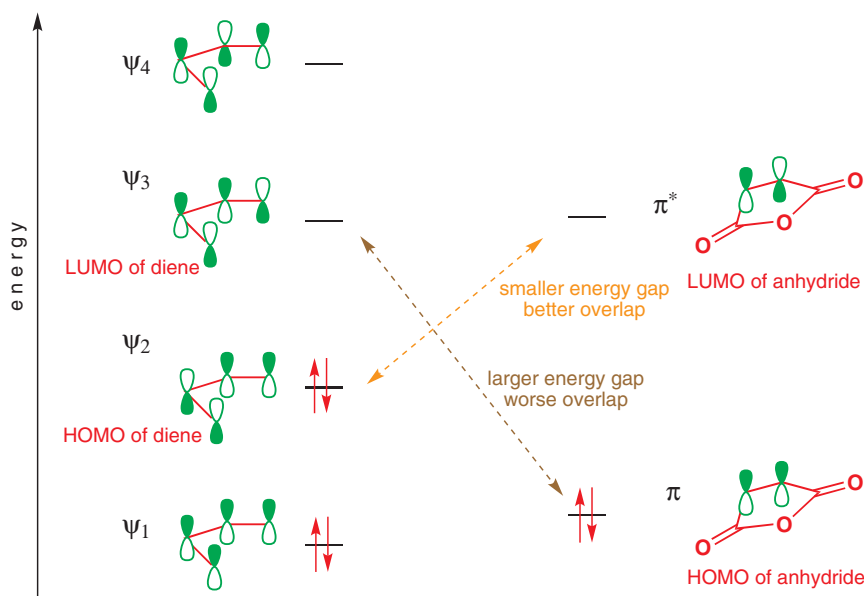
We shall now explain all this in one section using frontier molecular orbitals. Of all the kinds of organic reactions, pericyclic ones are the most tightly controlled by orbitals, and the development of the ideas we are about to expound is one of the greatest triumphs of modern theoretical chemistry. It is a beautiful and satisfying set of ideas based on very simple principles.

The frontier orbital description of cycloadditions

When an ionic cyclization reaction occurs, such as the lactonization at the head of this chapter, one important new bond is formed. It is enough to combine one full orbital with one empty orbital to make the new bond. But in a cycloaddition two new bonds are formed at the same time. We have to arrange for two filled p orbitals and two empty p orbitals to be available at the right place and with the right symmetry. See what happens if we draw the orbitals for the reaction above. We could try the HOMO (π) of the alkene and the LUMO (π^*) of the double bond in the anhydride (as in the margin). This combination is bonding at one end, but antibonding at the other so that no cycloaddition reaction occurs. It obviously doesn't help to use the other HOMO/LUMO pair, that is the HOMO of the alkene and the LUMO of the alkene, as they will have the same mismatched symmetry.

Now see what happens when we replace the alkene with a diene. We shall again use the LUMO of the electron-poor anhydride. Now the symmetry is right because there is a node in the middle of the HOMO of the diene (the HOMO is ψ_2 of the diene) just as there is in the LUMO of the dienophile.

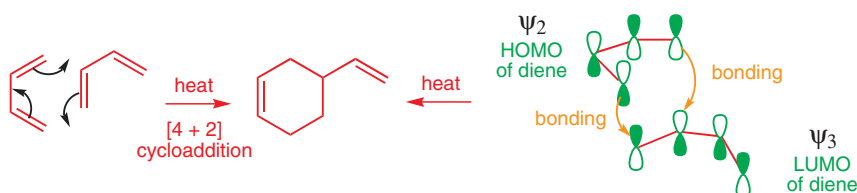
If we had tried the opposite arrangement, the LUMO of the diene (ψ_3) and the HOMO of the dienophile, the symmetry would again be right. The LUMO of the diene has two nodes and gives the same symmetry as the HOMO of the dienophile, which has no nodes. So either combination is excellent. In fact most Diels–Alder reactions use electron-deficient dienophiles and electron-rich dienes so we prefer the first arrangement. The electron-deficient dienophile has a low-energy LUMO and the electron-rich diene has a high-energy HOMO so this combination gives a better overlap in the transition state. The energy levels will look like this, and the interaction shown in orange is better than the interaction shown in brown because the orbitals are closer in energy.



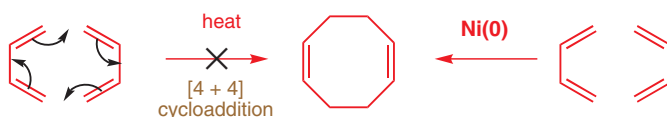
This is why we usually use dienophiles with conjugating groups for good Diels–Alder reactions. Dienes react rapidly with electrophiles because their HOMOs are relatively high in energy, but simple alkenes are not suitable electrophiles because they have relatively high energy LUMOs. The most effective modification we can make is to lower the energy of the alkene's LUMO by conjugating the double bond with an electron-withdrawing group such as carbonyl or nitro. These are the most common type of Diels–Alder reactions—between electron-rich dienes and electron-deficient dienophiles.

Dimerizations of dienes by cycloaddition reactions

Because dienes have relatively high-energy HOMOs and low-energy LUMOs they should be able to take part in cycloadditions with themselves. And indeed, dienes do dimerize, by a Diels–Alder reaction. One molecule of the diene plays the role of the dienophile. The symmetry is correct for the interaction shown, and we call such reactions (like all the Diels–Alder reactions in this chapter) '[4 + 2] cycloadditions'—the numbers referring to the number of atoms of each component taking part in the reaction.



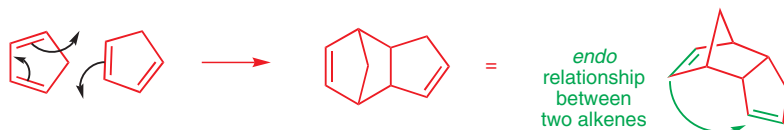
What dienes cannot do is form an eight-membered ring in one step in a [4 + 4] cycloaddition (although this is possible photochemically or with transition metal catalysis, as we shall see later).



You should have expected this failure because the ends of the required orbitals must again have the wrong symmetry, just as they had when we tried the alkene dimerization.

The orbital explanation for *endo* preference in Diels–Alder reactions

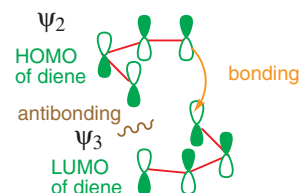
We are going to use a diene dimerization to add more detail to our explanation of the formation of *endo* products. To make matters even easier we shall look at the dimerization of a cyclic diene—we might almost say *the* cyclic diene—cyclopentadiene. We introduced the preference for *endo* products on p. 885 by saying there was a favourable electronic interaction between the conjugating group on the dienophile and the back of the diene.



If we now draw the frontier orbitals in the two components as they come together for the reaction, we can see first of all that the symmetry is correct for bond formation (orbitals shown in black). But we can also see what is happening at the back of the diene (orbitals in green). The symmetry of the orbitals is correct for a bonding interaction at the back of the diene too. This interaction does not lead to the formation of any new bonds but it leaves its imprint in the stereochemistry of the product. The *endo* product is favoured because of this bonding interaction across the space between the orbitals.

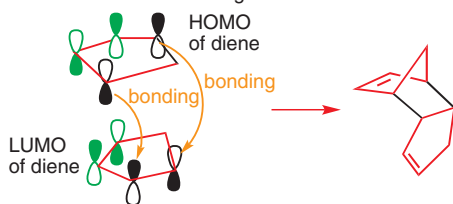
A rarer type is the **reverse electron demand Diels–Alder reaction** in which the dienophile has electron-donating groups and the diene has a conjugated electron-withdrawing group. These reactions use the HOMO of the dienophile and the LUMO of the diene. This combination still has the right orbital symmetry.

■ The same features of dienes allow them to react with both electrophiles and nucleophiles: see p. 148.

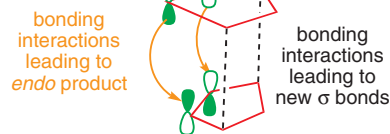


Interactive orbital explanation for *endo* preference in Diels–Alder reactions

orbital interactions leading to new bonds



orbital interactions favouring *endo* product

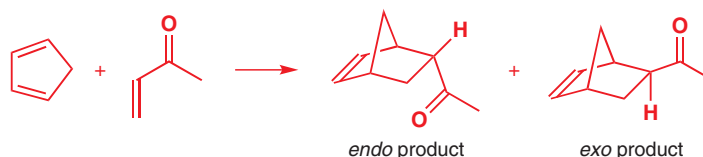


The solvent in the Diels–Alder reaction

We discussed some effects of varying the solvent in Chapter 12, and we shall now introduce a remarkable and useful special solvent effect in the Diels–Alder reaction. The reaction does not *need* a solvent and often the two reagents are just mixed together and heated. Solvents can be used but, because there are no ionic intermediates, it seems obvious that *which* solvent is unimportant—any solvent that simply dissolves both reagents will do. This is, in general, true and hydrocarbon solvents are often the best.

However, in the 1980s an extraordinary discovery was made. Water, a most unlikely solvent for most organic reactions, has a large accelerating effect on the Diels–Alder reaction. Even some water added to an organic solvent accelerates the reaction. And that is not all. The *endo* selectivity of these reactions is often superior to those in no solvent or in a hydrocarbon solvent. Here is a simple example.

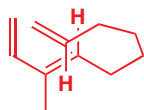
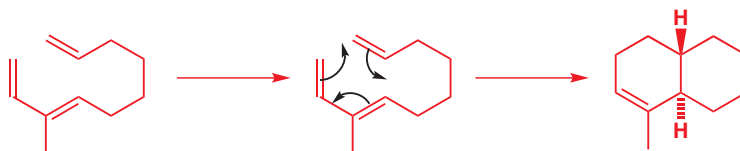
Solvent	Relative rate	<i>endo</i> : <i>exo</i> ratio
hydrocarbon (isooctane)	1	80:20
water	700	96:4



The suggestion is that the reagents, which are not soluble in water, are clumped together in oily drops by the water and forced into close proximity. Water is not exactly a solvent—it is almost an anti-solvent! Reactions like this are sometimes called reactions ‘on water’ rather than reactions ‘in water’.

Intramolecular Diels–Alder reactions

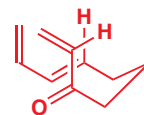
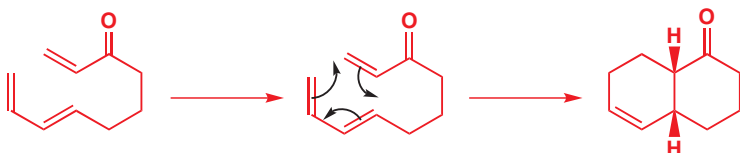
When the diene and the dienophile are already part of the same molecule it is not so important for them to be held together by bonding interactions across space and the *exo* product is often preferred. Indeed, many intramolecular Diels–Alder reactions are governed more by normal steric considerations than by the *endo* rule.



■ If you think about the way a Diels–Alder reaction goes, the forming ring must *always* adopt a boat-like conformation. This is clear if you make a model.

This reaction happens only because it is intramolecular. There is no conjugating group attached to the dienophile and so there are no orbitals to overlap with the back of the diene. The molecule simply folds up in the sterically most favourable way (as shown in the margin, with the linking chain adopting a chair-like conformation) and this leads to the *trans* ring junction. You can see this easily in the arrangement of the hydrogen atoms.

In the next example there is a carbonyl group conjugated with the dienophile. Now the less stable *cis* ring junction is formed because the molecule can fold so that the carbonyl group can enjoy a bonding overlap with the back of the diene. This time the linking chain has to adopt a boat-like conformation.

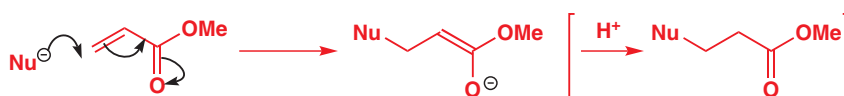


● Intramolecular Diels–Alder

Intramolecular Diels–Alder reactions may give the *endo* product or they may not! Be prepared for either *exo* or *endo* products or a mixture.

Regioselectivity in Diels–Alder reactions

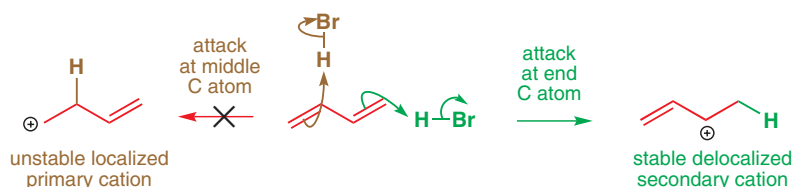
The compounds that we are now calling dienophiles were the stars of Chapters 22 and 25, where we called them **Michael acceptors** as they were the electrophilic partners in conjugate addition reactions. Nucleophiles always add to the β carbon atoms of these alkenes because the product is then a stable enolate. Ordinary alkenes do not react with nucleophiles.



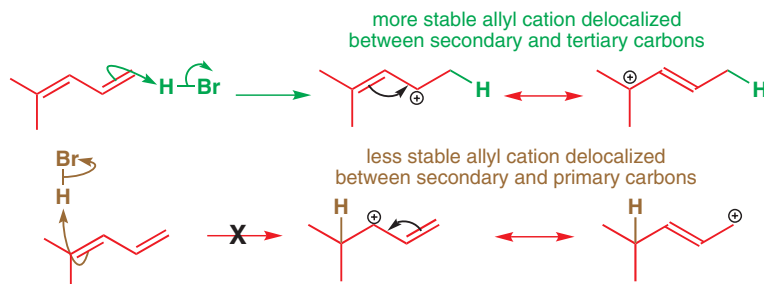
In frontier orbital terms this is because conjugation with a carbonyl group lowers the energy of the LUMO (the π^* orbital of the alkene) and at the same time distorts it so that the coefficient on the β carbon atom is larger than that on the α carbon atom. Nucleophiles approach the conjugated alkene along the axis of the large p orbital of the β carbon atom.

These same features can ensure regioselective Diels–Alder reactions. The same orbital of the dienophile is involved and, if the HOMO of the diene is also unsymmetrical, the regioselectivity of the reaction will be controlled by the two largest coefficients bonding together.

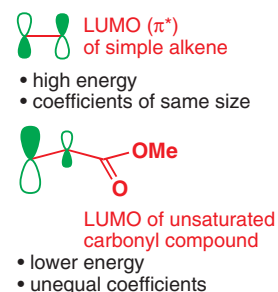
So what about distortion of the HOMO in the diene? If a diene reacts with an electrophile, the largest coefficient in the HOMO will direct the reaction. Consider the attack of HBr on a diene. We should expect attack at the ends of the diene because that gives the most stable cation—an allyl cation as an intermediate.

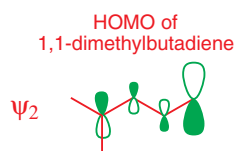


In orbital terms attack occurs at the ends of the diene because the coefficients in the HOMO are larger there. We need simply to look at the HOMO (ψ_2) of butadiene, shown in the margin, to see this. So it is not surprising that the dienes react in the Diels–Alder reaction through their end carbons. But supposing the two ends are different—which reacts now? We can again turn to the reaction with HBr as a guide. Addition of HBr to an unsymmetrical diene will give the more stable of the two possible allyl cations as the intermediate.



➡ This is discussed in Chapter 22.

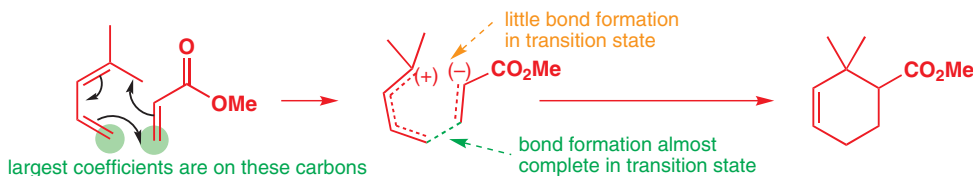




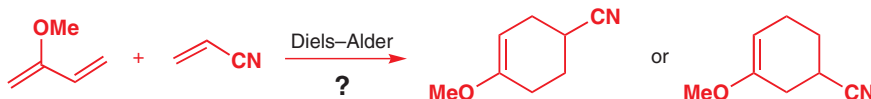
■ It is not 'cheating' to use the regioselectivity of chemical reactions to tell us about the coefficients in orbitals. Chemistry is about using experimental evidence to find out about the theoretical background and not about theory telling us what *ought* to happen. In fact, computational chemists have calculated the HOMO energies and coefficients of unsymmetrical dienes and have reached the same conclusions.

Interactive explanation of regioselectivity in Diels–Alder reactions

In orbital terms, this must mean that the HOMO of the diene is distorted so that the end that reacts has the larger coefficient. When the unsymmetrical diene and the unsymmetrical dienophile combine in a Diels–Alder reaction, the reaction itself becomes unsymmetrical. It remains concerted but, in the transition state, bond formation between the largest coefficients in each partner is more advanced and this determines the regioselectivity of the reaction.



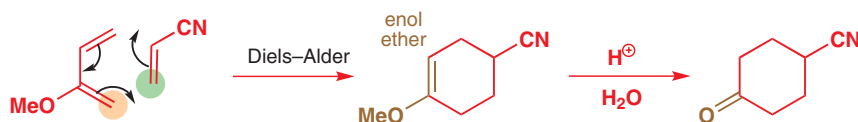
The simplest way to decide which product will be formed is to draw an 'ionic' stepwise mechanism for the reaction to establish which end of the diene will react with which end of the dienophile. Of course this stepwise mechanism is not completely correct but it does lead to the correct orientation of the reagents and you can draw the right mechanism afterwards. As an example, try a diene with a substituent in the middle. This is the reaction:



First decide where the diene will act as a nucleophile and where the dienophile will act as an electrophile. This indicates where the largest coefficients of the HOMO and LUMO must lie. The two circles represent those largest coefficients.



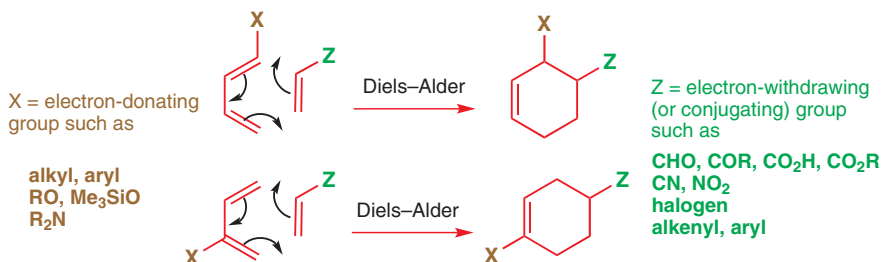
Now draw the reagents in the correct orientation for these two ends to combine and draw a concerted Diels–Alder reaction.



This is an important example because an enol ether functional group is present in the product, which can be hydrolysed to a ketone in aqueous acid (Chapter 20).

Summary of regioselectivity in Diels–Alder reactions

The important substitution patterns are a diene with an electron-donating group (X) at one end or in the middle and a dienophile with an electron-withdrawing group (Z) at one end. These are the products formed.



● **A useful mnemonic**

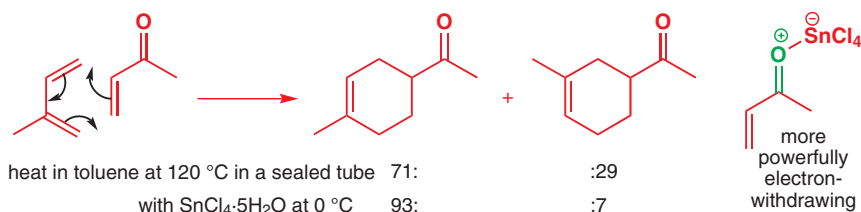
If you prefer a rule to remember, try this one.

- The Diels–Alder reaction is a cycloaddition with an aromatic transition state that is *ortho* and *para* directing.

You can see that this mnemonic works if you look at the two products above: the first has the two substituents X and Z on neighbouring carbon atoms, just like *ortho* substituents on a benzene ring, while the second has 1,4-related X and Z just like *para* substituents. The connection with aromaticity (the ‘aromatic transition state’) simply means that the transition state is cyclic and has six electrons. We have not yet explored the consequences of this, but we will do shortly.

Lewis acid catalysis in Diels–Alder reactions

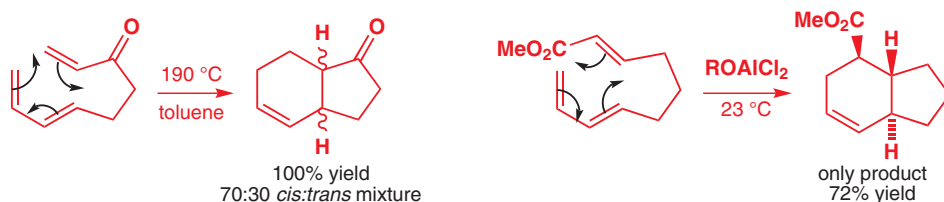
Where the reagents are unsymmetrical, a Lewis acid that can bind to the electron-withdrawing group of the dienophile often catalyses the reaction by lowering the LUMO of the dienophile still further. It has another important advantage: it increases the difference between the coefficients in the LUMO (a Lewis-acid complexed carbonyl group is a more powerful electron-withdrawing group) and may therefore increase regioselectivity.



This Diels–Alder reaction is useful because it produces a substitution pattern (*para*) common in natural terpenes (see Chapter 42). But the regioselectivity introduced by one methyl group on the diene is not very great—this reaction gives a 71:29 mixture when the two compounds are heated together at 120°C in a sealed tube. In the presence of the Lewis acid (SnCl_4) the reaction can be carried out at lower temperatures (below 25°C) without a sealed tube and the regioselectivity improves to 93:7.

Regioselectivity in intramolecular Diels–Alder reactions

Just as the stereoselectivity may be compromised in intramolecular reactions, so may the regioselectivity. It may be simply impossible for the reagents to get together in the ‘right’ orientation. The examples below have a very short chain—just three carbon atoms—joining diene to dienophile and so the same regioselectivity is found regardless of the position of the conjugating carbonyl group.



The first example has the ‘right’ orientation (*ortho*) but the second has the ‘wrong’ orientation (*meta*). The short tether entails no prospect of any other orientation and, as the reaction is intramolecular, it goes anyway. Notice the lower temperature required for the Lewis acid (ROAlCl_2) catalysed reaction.

The Woodward–Hoffmann description of the Diels–Alder reaction

Kenichi Fukui and Roald Hoffmann won the Nobel prize in 1981 (Woodward died in 1979 and so couldn't share this prize: he had already won a Nobel prize in 1965 for his work on synthesis) for the application of orbital symmetry to pericyclic reactions. There is an alternative description to the frontier orbital method we have used and you need to know a little about it. They started by considering a more fundamental correlation between the symmetry of all the orbitals in the starting materials and all the orbitals in the products. This is rather too complex for us to cover here, and we shall concentrate only on a summary of the conclusions—the **Woodward–Hoffmann rules**. The most important of these states:

● Woodward–Hoffmann rules

In a thermal pericyclic reaction the total number of $(4q + 2)_s$ and $(4r)_a$ components must be odd.

This needs some explanation. A component is a bond or orbital taking part in a pericyclic reaction as a single unit. A double bond is a π_2 component. The number 2 is the most important part of this designation and simply refers to the number of electrons. The prefix π tells us the type of electrons. A component may have any number of electrons (a diene is a π_4 component) but may not have mixtures of π and σ electrons. Now look back at the rule. Those designations $(4q + 2)$ and $(4r)$ simply refer to the number of electrons in the component where q and r are integers. An alkene is a π_2 component and so it is of the $(4q + 2)$ kind while a diene is a π_4 component and so is of the $(4r)$ kind. You have already seen the importance of $4n + 2$ numbers in aromaticity; here the significance is closely related.

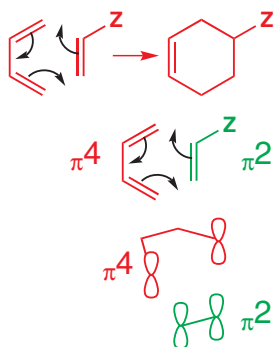
Now what about the suffixes 's' and 'a'? The suffix 's' stands for suprafacial and 'a' for antarafacial. A **suprafacial** component forms new bonds on the same face at both ends while an **antarafacial** component forms new bonds on opposite faces at both ends. If you find it easier to understand, you can think of the Woodward–Hoffmann rules like this:

● Woodward–Hoffmann rules: alternative version

In an allowed thermal pericyclic reaction this sum:

$$\begin{aligned} &\text{number of suprafacial components with 2, 6, or 10 electrons} \\ + &\text{number of antarafacial components with 0, 4, or 8 electrons} \\ = &\text{an odd number} \end{aligned}$$

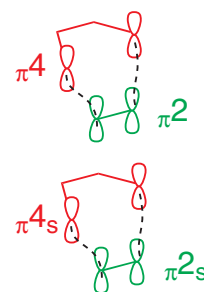
It's the number of relevant **components** that must be odd, not (obviously) the number of electrons, and you must ignore any components which aren't mentioned in the sum (for example you can have as many suprafacial components with four electrons as you like—they just don't count).



See how this works for the Diels–Alder reaction. Here is the routine.

1. Draw the mechanism for the reaction (we shall choose a general one).
2. Choose the components. All the bonds taking part in the mechanism must be included and no others.
3. Make a three-dimensional drawing of the way the components come together for the reaction, putting in orbitals at the ends of the components (only!). The orbitals are just unshaded p orbitals, and do not make up HOMOs or LUMOs nor any particular molecular orbital. Don't attempt to mix frontier orbital and Woodward–Hoffmann descriptions of pericyclic reactions.

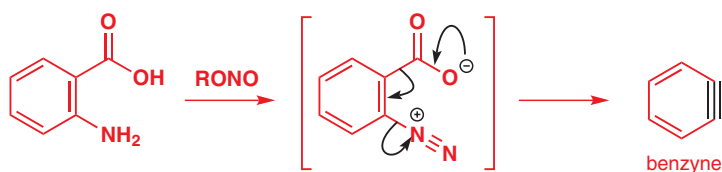
- Join up the components where new bonds are to be formed. Coloured dotted lines are often used.
- Label each component 's' or 'a' depending on whether new bonds are formed on the same or on opposite sides. In all of the cycloadditions you have seen so far (and indeed the vast majority of those you will ever see), both components react suprafacially.
- Count the number of $(4q + 2)_s$ and $(4r)_a$ components. If the total count is odd, the reaction is allowed. In this case, **there is one $(4q + 2)_s$ component (the alkene) and no $(4r)_a$ components. Total = 1 so it is an allowed reaction.** Components of the other symmetry, that is $(4q + 2)_a$ and $(4r)_s$ components, do not count. You can have as many of these as you want.



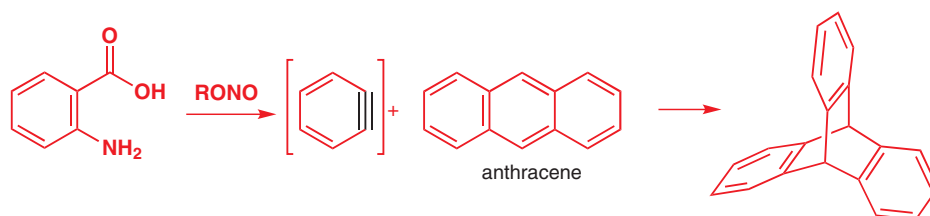
You may well feel that there is very little to be gained from the Woodward–Hoffmann treatment of the Diels–Alder reaction. It does not explain the *endo* selectivity nor the regioselectivity. However, the Woodward–Hoffmann treatment of other pericyclic reactions (particularly electrocyclic reactions, in the next chapter) is very helpful.

Trapping reactive intermediates by cycloadditions

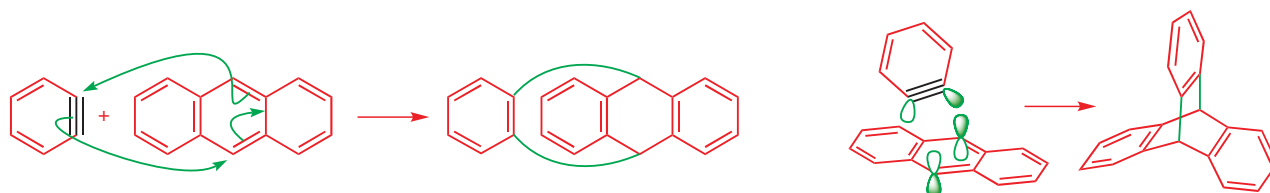
In Chapter 22 you met the remarkable intermediate benzyne. Convincing evidence for the existence of this implausible structure is provided by the fact that it can be trapped in a Diels–Alder reaction. One way of generating benzyne for this purpose is the diazotization of anthranilic acid (2-aminobenzoic acid).



Benzyne may not look like a good dienophile but it is an unstable electrophilic molecule so it must have a low-energy LUMO (π^* of the triple bond). If benzyne is generated in the presence of a diene, efficient Diels–Alder reactions take place. Anthracene gives a specially interesting product with a symmetrical cage structure.

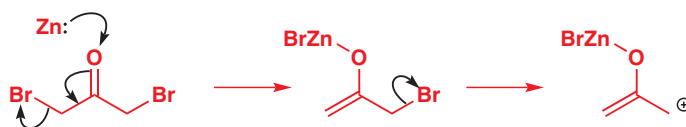


It is difficult to draw this mechanism convincingly. The two flat molecules approach each other in orthogonal planes, so that the (orbitals) of the localized π bond of benzyne interact with the p orbitals on the central ring of anthracene.

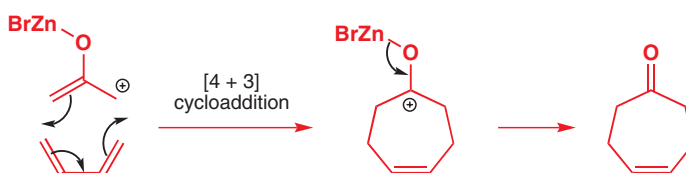


■ It also explains why the $[4 + 4]$ cycloaddition on p. 887 and the $[2 + 2]$ cycloaddition on p. 886 fail: draw out the reactions and you find there are *no* $(4q + 2)_s$ and $(4r)_a$ components—and you must have an *odd* number for a successful reaction.

Another intermediate for which a cycloaddition product provides convincing evidence is the oxyallyl cation. This compound can be made from α,α' -dibromoketones on treatment with zinc metal. The first step is the formation of a zinc enolate (compare the Reformatsky reaction), which can be drawn in terms of the attack of zinc on oxygen or bromine. Now the other bromine can leave as an anion. It could not do so before because it was next to an electron-withdrawing carbonyl group. Now it is next to an electron-rich enolate so the cation is stabilized by conjugation.



The allyl cation has three atoms but only two electrons so it can take part in cycloadditions with dienes—the total number of electrons is six, just as in the Diels–Alder reaction. This is a [4 + 3] all-suprafacial cycloaddition.

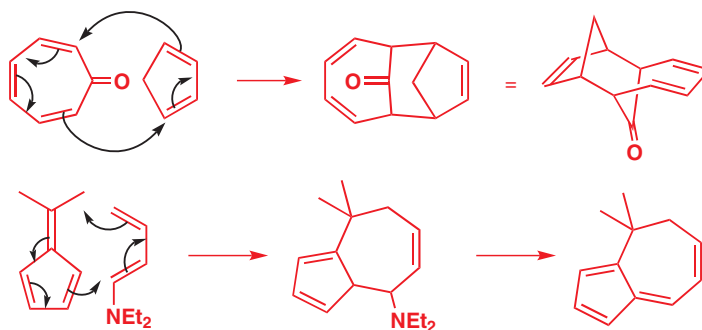


Interactive mechanism for [4 + 3] cycloaddition

Remember, the numbers in the brackets, [4 + 2] etc., refer to number of *atoms*. The numbers $(4q + 2)_s$ and $(4r)_a$ in the Woodward–Hoffman rules refer to the numbers of *electrons*. The [4 + 3] cycloaddition here still involves a $\pi 4_s$ and a $\pi 2_s$ component (i.e. it has one $(4q + 2)_s$ component and no $(4r)_a$ components, and is allowed).

Other thermal cycloadditions

A simple consequence of the Woodward–Hoffmann rules is that cycloadditions involving a total $(4n + 2)$ electrons, if they are all suprafacial, are always allowed: they must always involve an odd number of $(4q + 2)_s$ components. Such reactions are often referred to as having ‘aromatic transition states’ because of the obvious link with the aromatic requirement for $(4n + 2)$ electrons. Six is the most common $(4n + 2)$ number, but there are also a few cycloadditions involving ten electrons. These are mostly diene + triene, that is, $\pi 4_s + \pi 6_s$ cycloadditions. Here are a couple of examples.



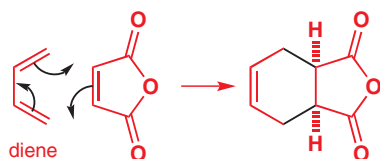
Interactive mechanism for [4 + 6] cycloaddition

In the first case, there is an *endo* relationship between the carbonyl group and the back of the diene—this product is formed in 100% yield. In the second case Et_2NH is lost from the first product under the reaction conditions to give the hydrocarbon shown. This type of reaction is more of an oddity: by far the most important type of cycloaddition is the Diels–Alder reaction.

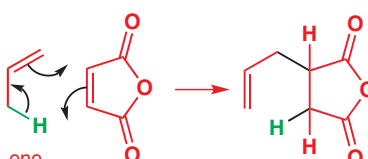
The Alder ‘ene’ reaction

The Diels–Alder reaction was originally called the ‘diene reaction’ so, when half of the famous team (Kurt Alder) discovered an analogous reaction that requires only one alkene, it was called the **Alder ene reaction** and the name has stuck. Compare here the Diels–Alder and the Alder ene reactions.

the Diels–Alder reaction



the Alder ene reaction



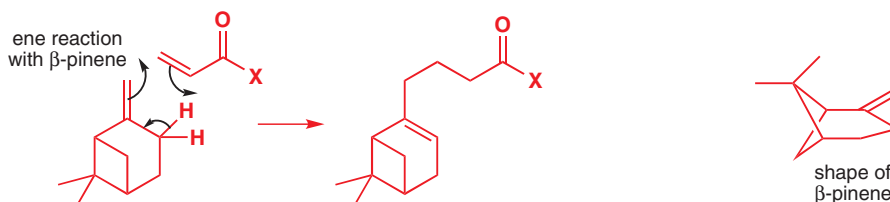
The simplest way to look at the ene reaction is to picture it as a Diels–Alder reaction in which one of the double bonds in the diene has been replaced by a C–H bond (green). The reaction does not form a new ring, the product has only one new C–C bond (shown in black on the product), and a hydrogen atom is transferred across space. Otherwise, the two reactions are remarkably similar.

The ene reaction is rather different in orbital terms. For the Woodward–Hoffmann description of the reaction we must use the two electrons of the C–H bond to replace the two electrons of the double bond in the Diels–Alder reaction, but we must make sure that all the orbitals are parallel, as shown.

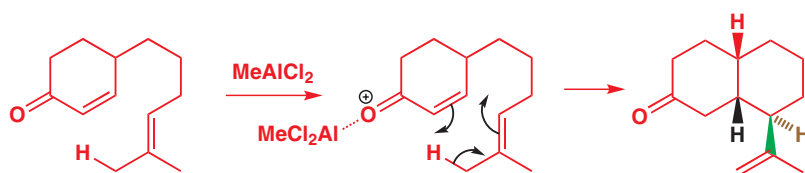
The C–H bond is parallel with the p orbitals of the ene so that the orbitals that overlap to form the new π bond are already parallel. The two molecules approach one another in parallel planes so that the orbitals that overlap to form the new σ bonds are already pointing towards each other. Because the electrons are of two types, π and σ , we must divide the ene into two components, one π^2 and one σ^2 . We can then have an all-suprafacial reaction with three components.

All three components are of the $(4q + 2)$, type so all count and the total is three—an odd number—so the reaction is allowed. We have skipped the step-by-step approach we used for the Diels–Alder reaction because the two are so similar, but you should convince yourself that you can apply it here.

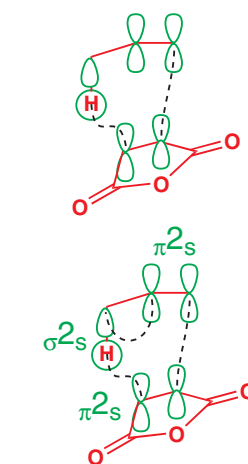
Now for some real examples. Most ene reactions with simple alkenes are with maleic anhydride. Other dienophiles—or enophiles as we should call them in this context—do not work very well. However, with one particular alkene, the natural pine tree terpene β -pinene, a reaction does occur with enophiles such as acrylates.



The major interaction between these two molecules is between the nucleophilic end of the exocyclic alkene and the electrophilic end of the acrylate. These atoms have the largest coefficients in the HOMO and LUMO, respectively, and, in the transition state, bond formation between these two will be more advanced than anywhere else. For most ordinary alkenes and enophiles, Lewis acid catalysis to make the enophile more electrophilic, or an intramolecular reaction (or both!), is necessary for an efficient ene reaction.

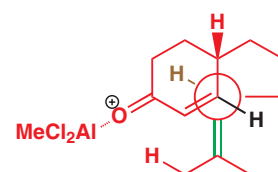


The 'ene' component is delivered to the bottom face of the enone, as its tether is too short for it to reach the top face, and a *cis* ring junction is formed. The stereochemistry of the third centre is most easily seen by a Newman projection (Chapter 16) of the reaction. In the diagram in the margin we are looking straight down the new C–C bond and the colour coding should help you to see how the stereochemistry follows.

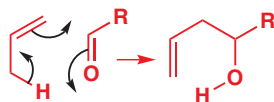


■ We discuss in more detail in Chapter 35 how to assign s or a with σ bonds. Here the σ bond reacts suprafacially because the 1s orbital of H has no nodes.

■ We looked at using tethers to constrain the formation of a single diastereoisomer in Chapter 32, p. 847.

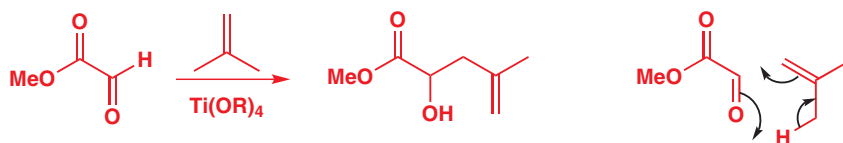


the carbonyl ene reaction



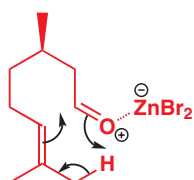
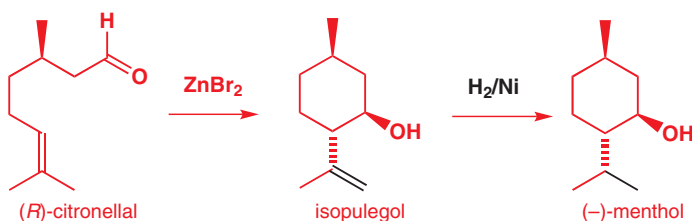
Since the twin roles of the enophile are to be attacked at one end by a C=C double bond and at the other by a proton, a carbonyl group is actually a very good enophile. These reactions are usually called ‘carbonyl ene’ reactions.

The important interaction is between the HOMO of the ene system and the LUMO of the carbonyl group—and a Lewis-acid catalyst can lower the energy of the LUMO still further. If there is a choice, the more electrophilic carbonyl group (the one with the lower LUMO) reacts.



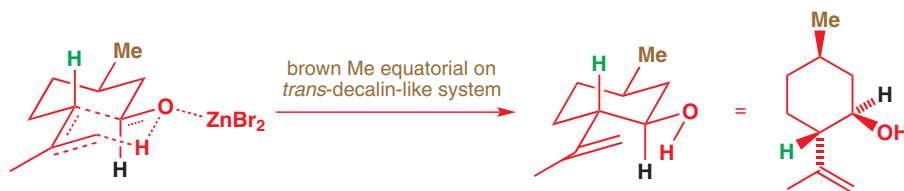
It may not be obvious that an ene reaction has occurred because of the symmetry of the alkene. The double bond in the product is not, in fact, in the same place as it was in the starting material, as the mechanism shows.

One carbonyl ene reaction is of commercial importance as it is part of a process for the production of menthol used to give a peppermint smell and taste to many products. This is an intramolecular ene reaction on another terpene derivative.



It is not obvious what has happened in the first step, but the movement of the alkene and the closure of the ring with the formation of one (not two) new C–C bonds should give you the clue that this is a Lewis-acid-catalysed carbonyl ene reaction.

The stereochemistry comes from an all-chair arrangement in the conformation of the transition state. The methyl group will adopt an equatorial position in this conformation, fixing the way the other bonds are formed. Again, colour coding should make it clearer what has happened.



Interactive mechanism for the intramolecular carbonyl ene reaction

Allowed reactions

Because a reaction is ‘allowed’ doesn’t mean that it will happen. It just means it is theoretically possible. In the same way you might be ‘allowed’ to jump off a three metre wall, but you wouldn’t do it.

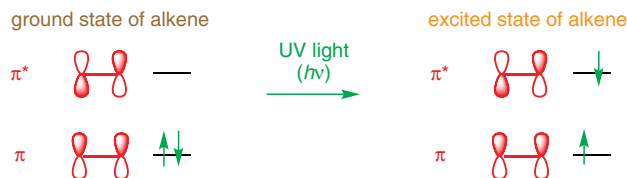
Menthol manufacture

It may seem odd to you to have a chemical process to produce menthol, which would be available naturally from mint plants. This process is now responsible for much of the world’s menthol production so it must make some sort of sense! The truth is that menthol *cultivation* is wasteful in good land that could produce food crops such as rice while the starting material for menthol *manufacture* is the same β -pinene we have just met. This is available in large quantities from pine trees grown on poor land for paper and furniture. The earlier stages of the process are discussed in Chapter 41.

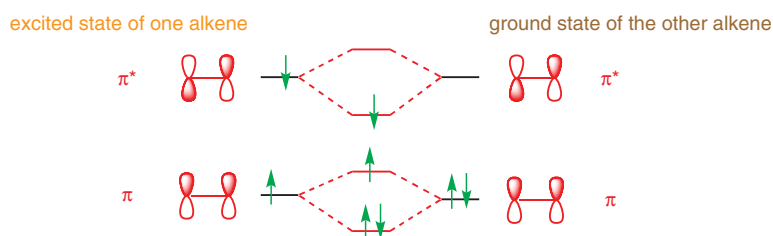
Photochemical [2 + 2] cycloadditions

We shall now leave six-electron cycloadditions such as the Diels–Alder and ene reactions and move on to some four-electron cycloadditions. Clearly, four is not a $(4n + 2)$ number, but when we described the Woodward–Hoffman rules on p. 892 we used the term ‘thermally’. All

suprafacial cycloadditions with $4n$ electrons *are* allowed if the reaction is not thermal (that is, driven by heat energy) but **photochemical** (that is, driven by light energy). Under photochemical conditions, the rules switch such that all the cycloadditions that are not allowed thermally are allowed photochemically. This works because the problem of the incompatible symmetry in trying to add two alkenes together is avoided by converting one of them into the excited state photochemically. First, one electron is excited by the light energy from the π to the π^* orbital.



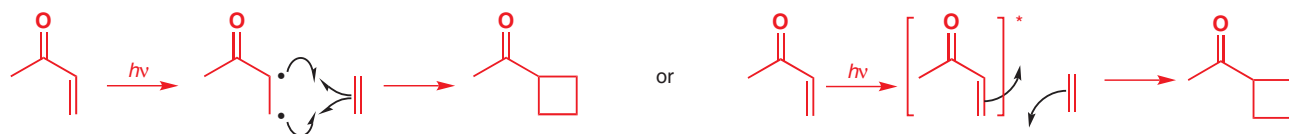
Now, combining the excited state of one alkene with the ground state of another solves the symmetry problem. Mixing the two π orbitals leads to two molecular orbitals, and two electrons go down in energy while only one goes up. Mixing the two π^* orbitals is as good—one electron goes down in energy and none goes up. The result is that three electrons go down in energy and only one goes up. Bonding can occur.



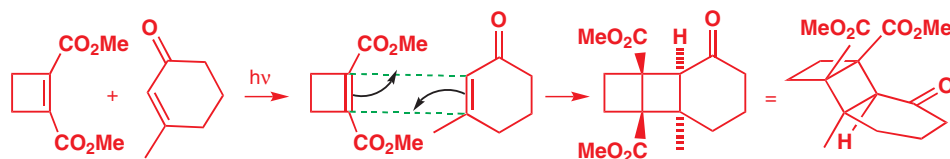
➔ In Chapter 7 we discussed why conjugated systems absorb UV light more readily than do unconjugated ones.

Alkenes can be dimerized photochemically in this way, but reaction between two different alkenes is more interesting. If one alkene is bonded to a conjugating group, it alone will absorb UV light and be excited while the other will remain in the ground state. It is difficult to draw a mechanism for these reactions as we have no simple way to represent the excited alkene. Some people draw it as a diradical (since each electron is in a different orbital); others prefer to write a concerted reaction on an excited alkene marked with an asterisk.

A photochemical [2 + 2] cycloaddition: two ways of writing the mechanism



The reaction is stereospecific within each component but there is no *endo* rule—there is a conjugating group but no ‘back of the diene’. The least hindered transition state usually results. The dotted lines on the central diagram simply show the bonds being formed. The two old rings keep out of each other’s way during the reaction and the conformation of the product looks reasonably unhindered.

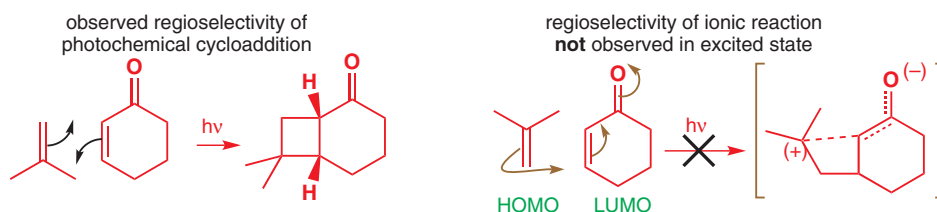


You may be wondering why the reaction works at all, given the strain in a four-membered ring: why doesn’t the product just go back to the two starting materials? This reverse reaction is governed by the Woodward–Hoffmann rules, just like the forward one, and to go back again the four-membered ring products would have to absorb light. But since they have now lost

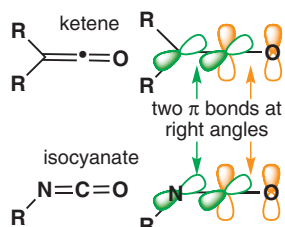
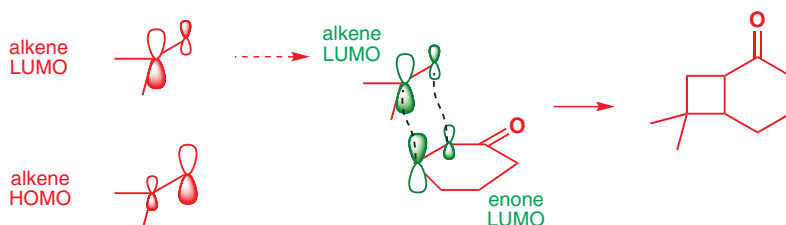
their π bonds they have no low-lying empty orbitals into which light can promote electrons (see Chapter 7). The reverse photochemical reaction is simply not possible because there is no mechanism for the compounds to absorb light.

Regioselectivity in photochemical [2 + 2] cycloadditions

The observed regioselectivity is shown below. If we had combined the HOMO of the alkene with the LUMO of the enone, as we should in a thermal reaction, we would expect the opposite orientation so as to use the larger coefficients of the frontier orbitals and to maximize charge stabilization in the transition state.



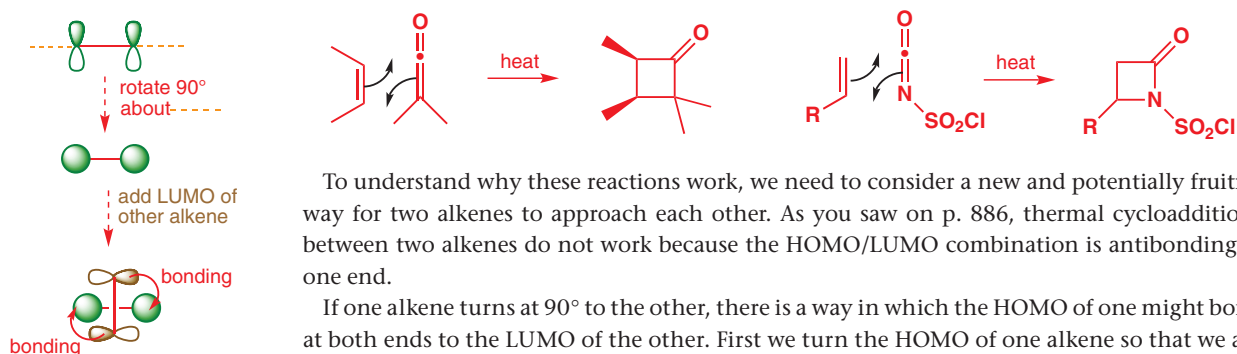
But we are not doing a thermal reaction. If you look back at the orbital diagram on p. 897, you will see that it is the HOMO/HOMO and LUMO/LUMO interactions that now matter in the reactions of the excited state. The sizes of the coefficients in the LUMO of the alkene are the other way round to those in the HOMO. There is one electron in this pair of orbitals—in the LUMO of the enone in fact, as the enone has been excited by the light—so overlap between the two LUMOs (shown in the frame) is bonding and leads to the observed product. The easiest way to work it out quickly is to draw the product you do *not* expect from a normal HOMO/LUMO or curly arrow controlled reaction.



Thermal [2 + 2] cycloadditions

Despite what we have told you about allowed cycloadditions, there *are* some thermal [2 + 2] cycloadditions giving four-membered rings. These feature a simple alkene reacting with an electrophilic alkene of a peculiar type. It must have two double bonds to the *same* carbon atom. The most important examples are ketenes and isocyanates. The structures have two π bonds at right angles.

Here are typical reactions of dimethyl ketene to give a cyclobutanone and chlorosulfonyl isocyanate to give a β -lactam.



To understand why these reactions work, we need to consider a new and potentially fruitful way for two alkenes to approach each other. As you saw on p. 886, thermal cycloadditions between two alkenes do not work because the HOMO/LUMO combination is antibonding at one end.

If one alkene turns at 90° to the other, there is a way in which the HOMO of one might bond at both ends to the LUMO of the other. First we turn the HOMO of one alkene so that we are

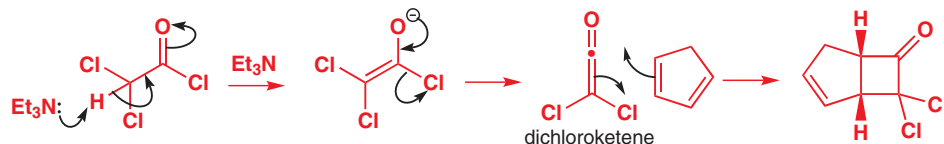
looking down on the p orbitals. Then we add the LUMO of the other alkene on top of this HOMO and at 90° to it so that there is the possibility of bonding overlap at both ends.

This arrangement looks quite promising until we notice that there is antibonding at the other two corners! Overall there is no net bonding. We can tilt the balance in favour of bonding by adding a p orbital to one end of the LUMO and at a right angle to it so that both orbitals of the HOMO can bond to this extra p orbital. There are now four bonding interactions but only two antibonding. The balance is in favour of a reaction. This is also quite difficult to draw!

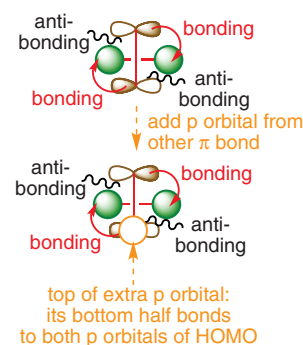
Ketenes have a central sp carbon atom with an extra π bond (the C=O) at right angles to the first alkene—perfect for thermal [2 + 2] cycloadditions. They are also electrophilic and so have suitable low-energy LUMOs.

Ketene [2 + 2] cycloadditions

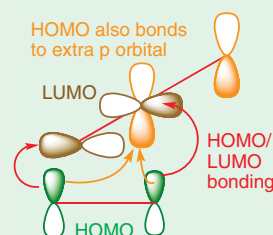
Ketene itself is usually made by high-temperature pyrolysis of acetone but some ketenes are easily made in solution. The very acidic proton on dichloroacetyl chloride can be removed even with a tertiary amine and loss of chloride ion then gives dichloroketene in an E1cB elimination reaction. If the elimination is carried out in the presence of cyclopentadiene a very efficient regio- and stereospecific [2 + 2] cycloaddition occurs.



The most nucleophilic atom on the diene adds to the most electrophilic atom on the ketene and the *cis* geometry at the ring junction comes from the *cis* double bond of cyclopentadiene. It is impressive that even this excellent diene undergoes no Diels–Alder reaction with ketene as dienophile. The [2 + 2] cycloaddition must be much faster.

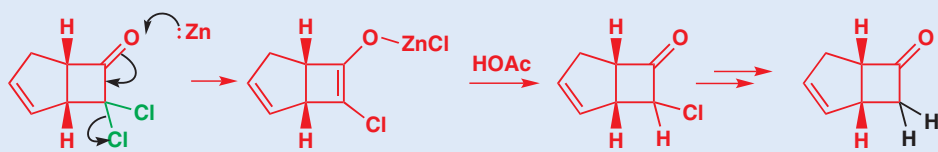


■ If you find the drawing above difficult to understand, try a three-dimensional representation.

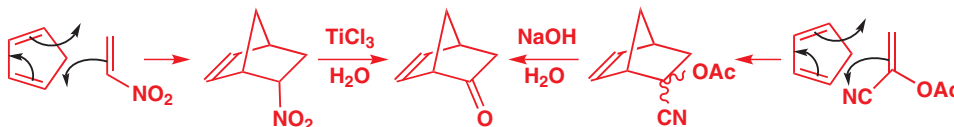


Using the products

Dichloroketene is convenient to use, but the two chlorine atoms are not usually needed in the product. Fortunately, these can be removed by zinc metal in acetic acid solution. Zinc forms a zinc enolate, which is converted into the ketone by the acid. Repetition removes both chlorine atoms. You saw the reductive formation of a zinc enolate earlier in the chapter (p. 894) and in the Reformatsky reaction (Chapter 26, p. 631).



But what do we do if we *want* the product of a ketene [4 + 2] cycloaddition? We must use a compound that is not a ketene but that can be transformed into a ketone afterwards—a **masked ketene** or a **ketene equivalent**. The two most important types are nitroalkenes and compounds such as the 'cyanohydrin ester' in the second example.



Finding the starting materials for a cyclobutanone synthesis

The disconnection of a four-membered ring is very simple—you just split in half and draw the two alkenes. There may be two ways to do this.

■ The conversion of nitro compounds to ketones by TiCl_3 is an alternative to the Nef reaction that you met in Chapter 26 (p. 631), and you should be able to write a mechanism for the reaction involving NaOH yourself.



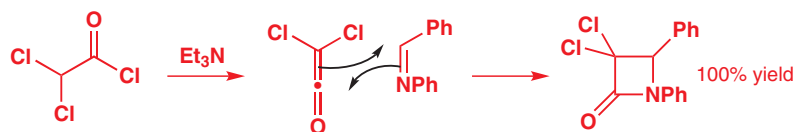
Both sets of starting materials look all right—the regiochemistry is correct for the first and doesn't matter for the second. However, we prefer the second because we can control the stereochemistry by using *cis*-butene as the alkene and we can make the reaction work better by using dichloroketene instead of ketene itself, reducing out the chlorine atoms with zinc.

Synthesis of β -lactams by [2 + 2] cycloadditions

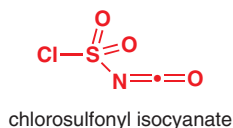
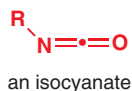
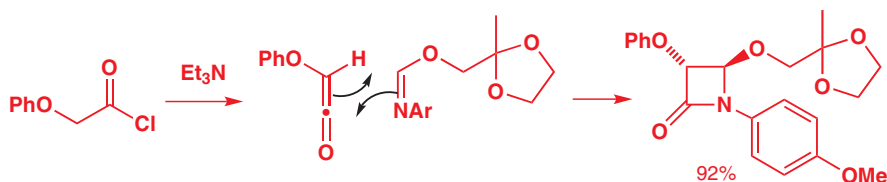
Now the disconnections are really different—one requires addition of a ketene to an imine and the other the addition of an isocyanate to an alkene. Isocyanates are like ketenes, but have a nitrogen atom instead of the end carbon atom. Otherwise the orbitals are the same.



And the good news is that both work, providing we have the right substituents on nitrogen. The dichloroacetyl chloride trick works well with imines and, as you ought to expect, the more nucleophilic nitrogen atom attacks the carbonyl group of the ketene so that the regioselectivity is right to make β -lactams.

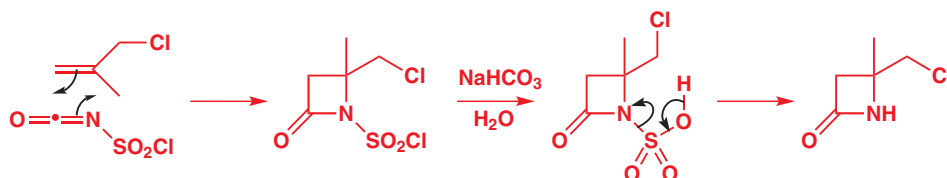


If both components have one substituent, these will end up *trans* on the four-membered ring just to keep out of each other's way. This example has more functionality and the product is used to make β -lactams with antibiotic activity.



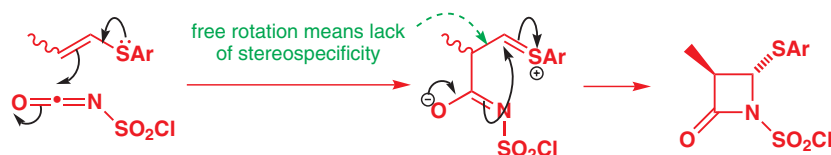
You will notice that in both of these examples there is an aryl substituent on the nitrogen atom of the imine. This is simply because *N*-aryl imines are more stable than their *NH* analogues (Chapter 11, p. 231).

When we wish to make β -lactams by the alternative addition of an isocyanate to an alkene, a substituent on nitrogen is again required, but for quite a different reason. Because alkenes are only moderately nucleophilic, we need a strongly electron-withdrawing group on the isocyanate that can be removed after the cycloaddition, and the most popular by far is the chlorosulfonyl group. The main reason for its popularity is the commercial availability of chlorosulfonyl isocyanate. It reacts even with simple alkenes.



The alkene's HOMO interacts with the isocyanate's LUMO, and the most electrophilic atom is the carbonyl carbon so this is where the terminal carbon atom of the alkene attacks. The chlorosulfonyl group can be removed simply by hydrolysis under mild conditions via the sulfonic acid.

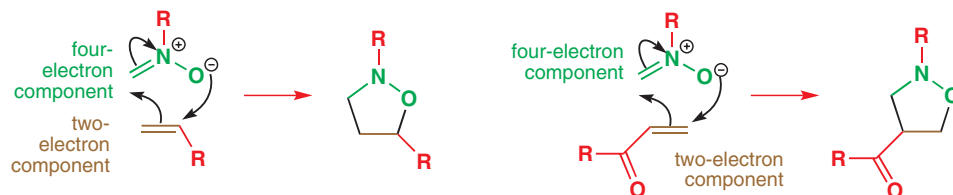
With a more electron-rich alkene—an enol ether, for example, or the following example with its sulfur analogue, a vinyl sulfide—the reaction ceases to be a concerted process and occurs stepwise. We know this must be the case in the next example because, even though the starting material is an *E/Z* mixture, the product has only *trans* stereochemistry: it is stereoselective rather than stereospecific, indicating the presence of an intermediate in which free rotation can take place.



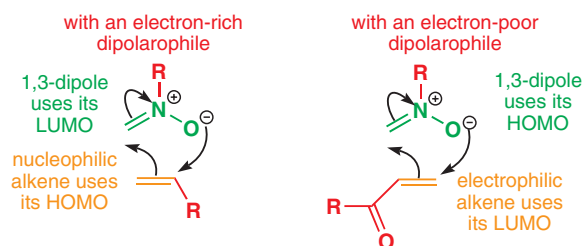
■ The lack of stereospecificity in some non-concerted reactions is discussed in Chapter 38 in relation to carbenes.

Making five-membered rings: 1,3-dipolar cycloadditions

We have seen how to make four-membered rings by [2 + 2] cycloadditions, how to make six-membered rings by [4 + 2] cycloadditions, and an example of making a seven-membered ring by a [4 + 3] cycloaddition. But what about five-membered rings? What we need is a three-atom, four-electron equivalent of a 'diene' and we can do a Diels–Alder reaction. Such molecules exist: they are called 1,3-dipoles and they are good reagents for [3 + 2] cycloadditions. The molecule containing N and O atoms labelled 'four-electron component' is an example. It has a nucleophilic end (O^-) and an electrophilic end—the end of the double bond next to the central N^+ . These are 1,3-related, so it is indeed a 1,3-dipole.

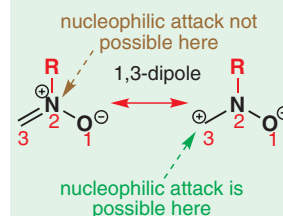


This functional group is known as a nitron. You could think of it as the *N*-oxide of an imine. The nitron gets its four electrons in this way: there are two π electrons in the $\text{N}=\text{C}$ double bond and the other two come from one of the lone pairs on the oxygen atom. The two-electron component in each of these reactions is an alkene which, in a Diels–Alder reaction, would be called a dienophile. Here it is called a **dipolarophile**. Simple alkenes (which are bad dienophiles) are good dipolarophiles and so are electron-deficient alkenes. The difference between dienes and 1,3-dipoles is that dienes are nucleophilic and prefer to use their HOMO in cycloadditions with electron-deficient dienophiles while 1,3-dipoles, as their name implies, are both electrophilic and nucleophilic. They can use either their HOMO or their LUMO depending on whether the dipolarophile is electron-deficient or electron-rich.



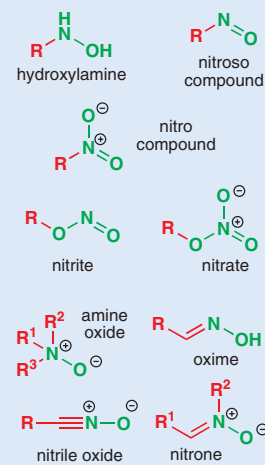
➔ You saw 1,3-dipolar cycloaddition being used to make heterocycles in Chapter 30 (pp. 772–775).

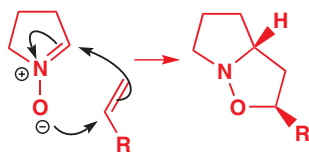
■ The charges make the nitron look like a 1,2-dipole, but nucleophilic attack on N^+ is impossible.



N–O functionality

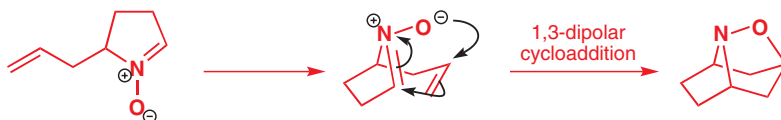
There are many functional groups containing N–O bonds. Here are a few:





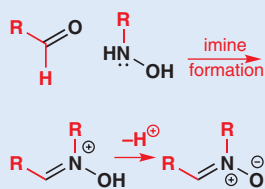
One important nitronium is a cyclic compound that has the structure in the margin and adds to dipolarophiles (essentially any alkene) in a [3 + 2] cycloaddition to give two five-membered rings fused together. The stereochemistry comes from the best approach with the least steric hindrance, as shown. There is no *endo* rule in these cycloadditions as there is no conjugating group to interact across space at the back of the dipole or dipolarophile. The product shown here is the more stable *exo* product.

If the alkene is already joined on to the nitronium by a covalent bond, the dipolar cycloaddition is an intramolecular reaction, and one particular outcome may be dictated by the impossibility of the alternatives. In the simple case below, the product has a beautifully symmetrical cage structure. The mechanism shows the only way in which the molecule can fold up to allow a 1,3-dipolar cycloaddition to occur.



Making nitrones

The most important route to nitrones starts from hydroxylamines. Open-chain nitrones are usually made simply by imine formation between a hydroxylamine and an aldehyde.



The importance of the Diels–Alder reaction is that it makes six-membered rings with control over stereochemistry. The importance of 1,3-dipolar cycloadditions is not so much in the heterocyclic products but in what can be done with them. Almost always, the first formed heterocyclic ring is broken down in some way by carefully controlled reactions. The nitronium adducts we have just seen contain a weak N–O single bond that can be selectively cleaved by reduction. Reagents such as LiAlH_4 or zinc metal in various solvents (acetic acid is popular) or hydrogenation over catalysts such as nickel reduce the N–O bond to give NH and OH functionality without changing the structure or stereochemistry of the rest of the molecule. From the examples above, we get these products:

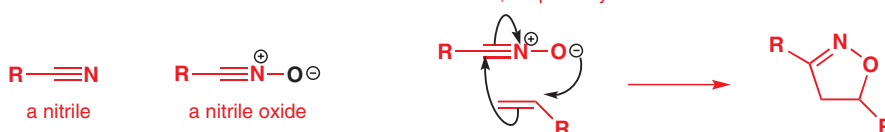


In each cycloaddition, one permanent C–C and one C–O bond (shown in brown) were made. These were retained while the N–O bond present in the original dipole was discarded. The final product is an amino-alcohol with a 1,3-relationship between the OH and NH groups.

Linear 1,3-dipoles

In the Diels–Alder reaction, the dienes had to have an *s-cis* conformation about the central single bond so that they were already in the shape of the product. Many useful 1,3-dipoles are actually linear and although their 1,3-dipolar cycloadditions look very awkward they still work well. We shall start with the nitrile oxides, which have a triple bond where the nitronium had a double bond.

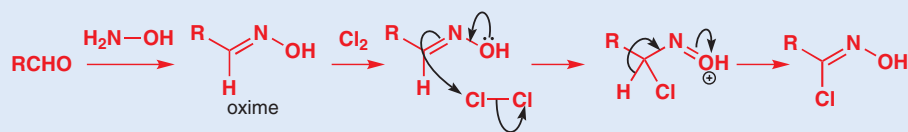
a 1,3-dipolar cycloaddition with a nitrile oxide



Interactive mechanism for nitrile oxide cycloaddition

Making nitrile oxides

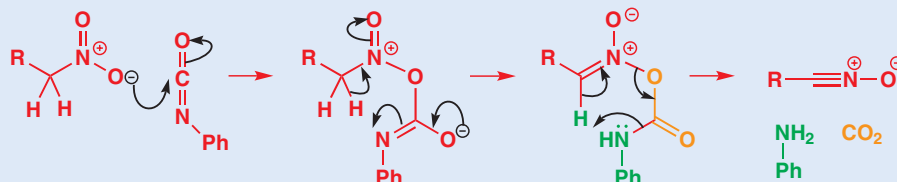
There are two important routes to these compounds, both of which feature interesting chemistry. Oximes, easily made from aldehydes with hydroxylamine ($\text{NH}_2\text{--OH}$), are rather enol-like and can be chlorinated on carbon.



Treatment of the chloro-oxime with base (Et_3N is strong enough) leads directly to the nitrile oxide with the loss of HCl . This is an elimination of a curious kind as we cannot draw a connected chain of arrows for it. We must use two steps—removal of the OH proton and then loss of chloride. It is a γ elimination rather than the more common β elimination.



The other method starts from nitroalkanes and is a dehydration. Inspect the two molecules and you will see that the nitro compound contains one molecule of H_2O more than the nitrile oxide. But how to remove the molecule of water? The reagent usually chosen is phenyl isocyanate ($\text{Ph-N}=\text{C}=\text{O}$), which removes the molecule of water atom-by-atom to give aniline (PhNH_2) and CO_2 . This is probably the mechanism, although the last step might not be concerted, as we have shown.

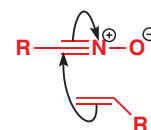


Interactive mechanism for nitrile oxide formation

As you might expect, this $[3+2]$ cycloaddition is a reaction involving the HOMO of the alkene and the LUMO of the nitrile oxide so that the leading interaction that determines the structure of the product is the one in the margin. If there is stereochemistry in the alkene, it is faithfully reproduced in the heterocyclic adduct as is usual for a concerted cycloaddition.



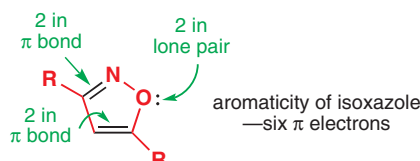
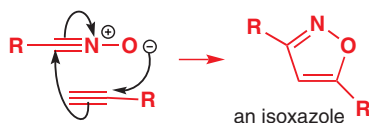
LUMO of nitrile oxide



HOMO of alkene

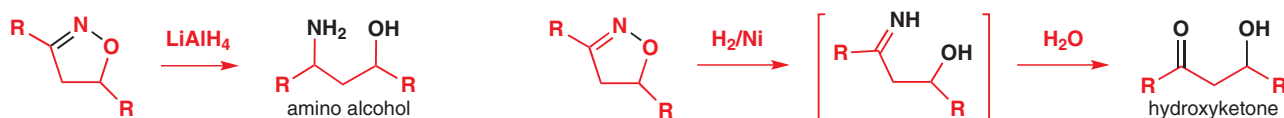
Both partners in nitrile oxide cycloadditions can have triple bonds—the product is then a stable aromatic heterocycle called an isoxazole.

cycloaddition of nitrile oxide and alkyne



Interactive mechanism for isoxazole formation

Reduction of the $\text{N}-\text{O}$ bond and the $\text{C}=\text{N}$ double bond of the nitrile oxide cycloadducts produces useful amino alcohols with a 1,3-relationship between the two functional groups. As the $\text{N}-\text{O}$ bond is the weaker of the two, it is alternatively possible to reduce just that and leave the $\text{C}=\text{N}$ bond alone. This gives an imine, which usually hydrolyses during work-up.

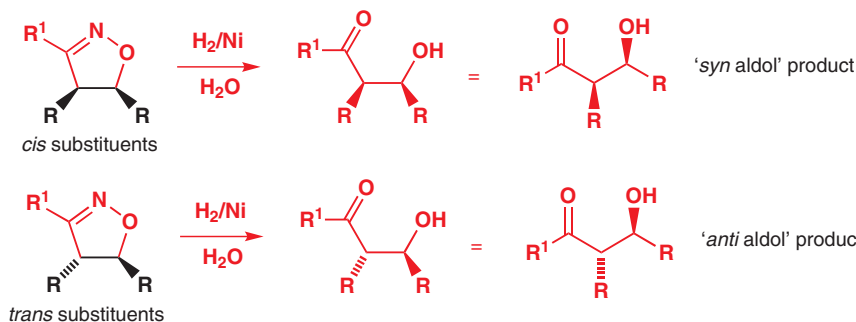
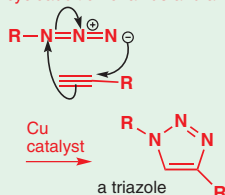


Any stereochemistry in the adduct is preserved right through this reduction and hydrolysis sequence: you might like to compare the products with the products of the stereoselective aldol reactions you saw in Chapter 33.

Other heterocycles by 1,3-dipolar cycloaddition

The synthesis of aromatic heterocycles by 1,3-dipolar cycloadditions was also treated in some detail in Chapter 30. There we discussed the important related reaction of azides with alkynes to make triazoles (p. 774).

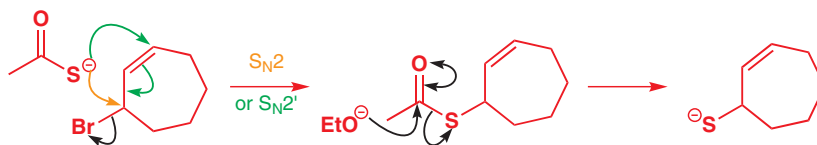
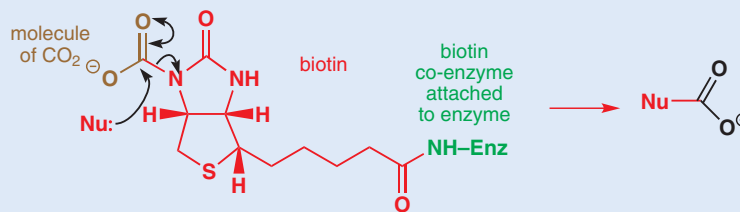
cycloaddition of azide and alkyne



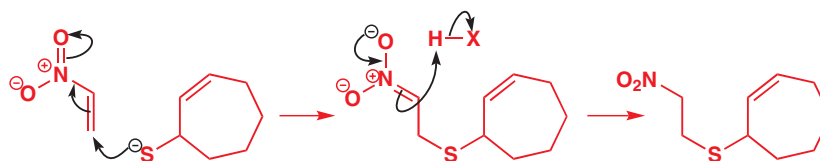
We shall end this section with the illustration of a beautiful intramolecular 1,3-dipolar cycloaddition that was used in the synthesis of the vitamin biotin. Starting at the beginning of the synthesis will allow you to revise some reactions from earlier chapters. The starting material is a simple cyclic allylic bromide that undergoes an efficient S_N2 reaction with a sulfur nucleophile. In fact, we don’t know (or care!) whether this is an S_N2 or S_N2' reaction as the product of both reactions is the same. This sort of chemistry was discussed in Chapter 24 if you need to check up on it. Notice that it is the sulfur atom that does the attack—it is the soft end of the nucleophile and better at S_N2 reactions. The next step is the cleavage of the ester group to reveal the thiolate anion.

Biotin

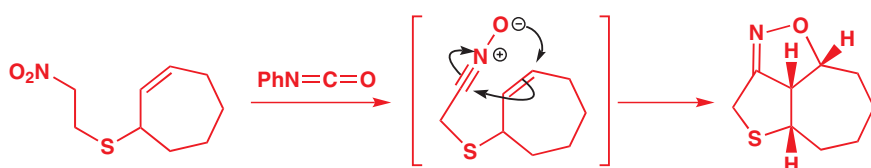
Biotin is an enzyme cofactor that activates and transports CO_2 for use as an electrophile in biochemical reactions.



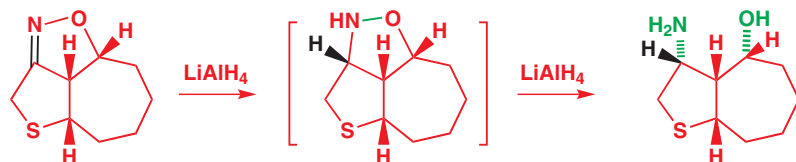
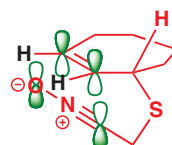
The nucleophilic thiolate anion does a conjugate addition (Chapter 22) on to a nitroalkene.



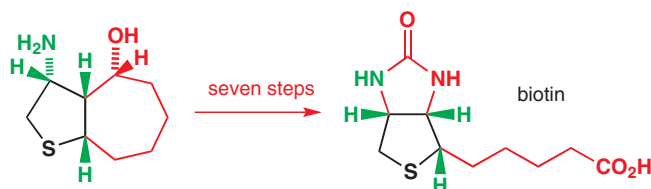
Now comes the exciting moment. The nitroalkene gives the nitrile oxide directly on dehydration with $\text{PhN}=\text{C}=\text{O}$ and the cycloaddition occurs spontaneously in the only way it can, given the intramolecular nature of the reaction.



In the margin we show how this reaction works—the nitrile oxide comes up from the underside of the seven-membered ring, pushing the black hydrogen atoms upwards and making all the rings join up in a *cis* fashion. Next the cycloadduct is reduced completely with LiAlH_4 so that both the N–O and C=N bonds are cleaved. This step is very stereoselective so the C=N reduction probably precedes the N–O cleavage and the hydride has to attack from the outside (top) face of the molecule. These considerations are explored more thoroughly in Chapter 32.



The sulfur-containing ring and the stereochemistry of biotin are already defined. In the seven steps that follow, the rest of the molecule is assembled. The most important is the breaking open of the seven-membered ring by a Beckmann rearrangement (which you will meet in Chapter 36).

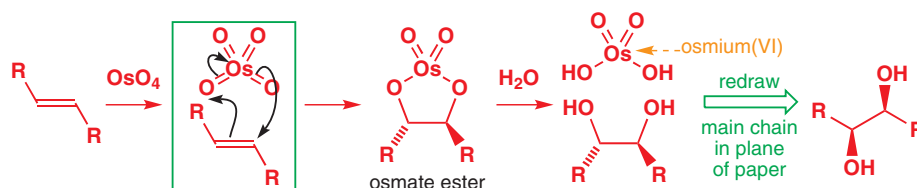


Two very important synthetic reactions: cycloaddition of alkenes with osmium tetroxide and with ozone

We shall end this chapter with two very important reactions, both of which we have alluded to earlier in the book (Chapter 19). These reactions are very important not just because of their mechanisms, which you must be aware of, but even more because of their usefulness in synthetic chemistry, and in that regard they are second only to the Diels–Alder reaction when considering all the reactions in this chapter. They are both oxidations—one involves osmium tetroxide (OsO_4) and one involves ozone (O_3) and they both involve cycloaddition.

OsO_4 adds two hydroxyl groups *syn* to a double bond

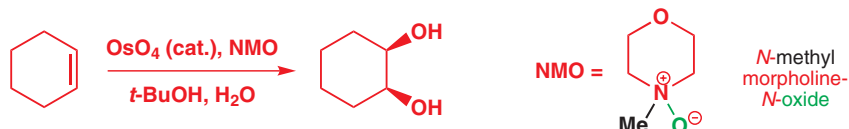
In Chapter 19 we emphasized the stereospecificity of this reaction but now we want to consider the nature of the first step (in the green frame). This is a cycloaddition between the osmium tetroxide and the alkene. You can treat the OsO_4 like a dipole, although it isn't drawn as one because osmium has plenty of orbitals to accommodate four double bonds. The reaction is a $[3 + 2]$ cycloaddition or a 1,3-dipolar cycloaddition, whichever you prefer.



Interactive mechanism for dihydroxylation of alkenes

The osmate ester isn't the required product, and the reaction is usually done in the presence of water (the usual solvent is a *t*-BuOH-water mixture), which hydrolyses the osmate ester to the diol. Because both oxygen atoms were added in one concerted step during the cycloaddition, their relative stereochemistry must remain *syn*.

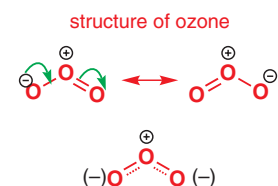
Note that, in the cycloaddition, one arrow stops on osmium and another starts on the other side. Osmium therefore gains a lone pair of electrons and is reduced from Os(VIII) to Os(VI)—the reaction is therefore an oxidation, and it's one that is very specific to C=C double bonds (as we mentioned in Chapter 23). As written, it would involve a whole equivalent of the expensive, toxic, and heavy metal osmium, but it can be made catalytic by introducing a reagent to oxidize Os(VI) back to Os(VIII). The usual reagent is *N*-methylmorpholine-*N*-oxide (NMO) or Fe(III), and typical conditions for an osmylation, or dihydroxylation, reaction are shown in the scheme below.



In behaviour that is typical of a 1,3-dipolar cycloaddition reaction, OsO₄ reacts almost as well with electron-poor as with electron-rich alkenes. OsO₄ simply chooses to attack the alkene HOMO or its LUMO, depending on which gives the best interaction. This is quite different from the electrophilic addition of *m*-CPBA or Br₂ to alkenes.

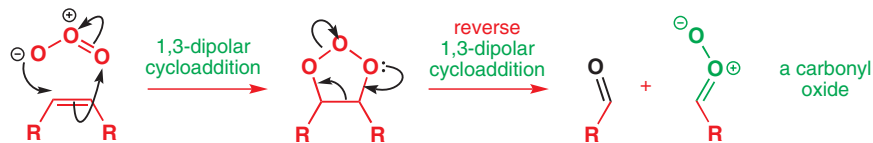
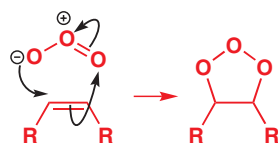


A cycloaddition that destroys bonds: ozonolysis

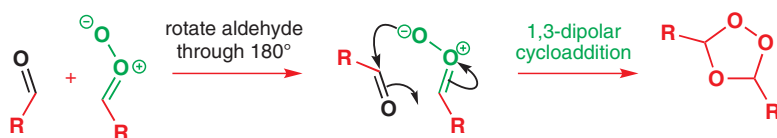


Our last type of cycloaddition is most unusual. It starts as a 1,3-dipolar cycloaddition but eventually becomes a method of cleaving π bonds in an oxidative fashion so that they end up as two carbonyl groups. The reagent is ozone, O₃. Again, you met this reaction in Chapter 19, but we can now show you the full, remarkable details of the reaction mechanism.

Ozone is a symmetrical bent molecule with a central positively charged oxygen atom and two terminal oxygen atoms that share a negative charge. It is a 1,3-dipole and does typical 1,3-dipolar cycloadditions with alkenes. The product is a very unstable compound. The O–O single bond (bond energy 140 kJ mol^{−1}) is a very weak bond—much weaker than the N–O bond (180 kJ mol^{−1}) we have been describing as weak in previous examples—and this heterocycle has two of them. It immediately decomposes—by a *reverse* 1,3-dipolar cycloaddition.

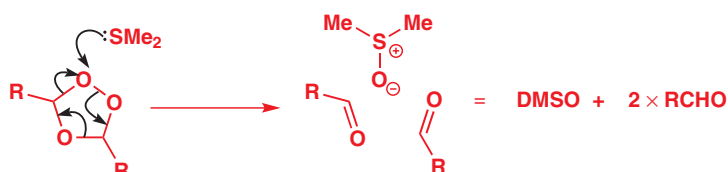


The products are a simple aldehyde on the left and a new, rather unstable looking molecule—a 1,3-dipole known as a carbonyl oxide—on the right. At least it no longer has any true O–O single bonds (the one that looks like a single bond is part of a delocalized system like the one in ozone). Being a 1,3-dipole, it now adds to the aldehyde in a third cycloaddition step. It might just add back the way it came, but it much prefers to add in the other way round, with the nucleophilic oxyanion attacking the carbon atom of the carbonyl group like this.



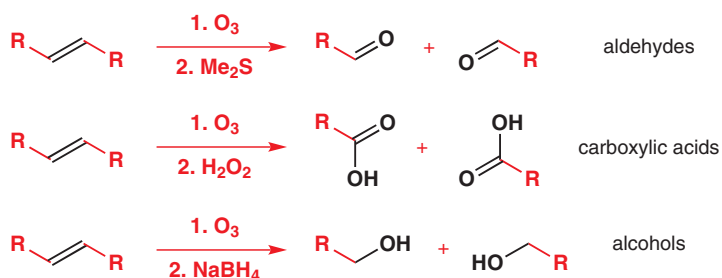
 Interactive mechanism of ozonolysis

This compound—known as an ozonide—is the first stable product of the reaction with ozone. It is the culmination of two 1,3-dipolar cycloadditions and one reverse 1,3-dipolar cycloaddition. It is still not that stable and is quite explosive, so for the reaction to be of any use it needs decomposing. The way this is usually done is with dimethylsulfide or Ph_3P , which attacks the ozonide to give DMSO and two molecules of aldehyde.



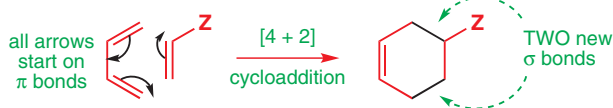
The ozonide will also react with oxidizing agents such as H_2O_2 to give carboxylic acids, or with more powerful reducing agents such as NaBH_4 to give alcohols. Here are the overall transformations—each cleaves a double bond—it is called an ozonolysis.

ozonolysis of alkenes to...



Summary of cycloaddition reactions

- A cycloaddition is a one-step ring-forming reaction between two conjugated π systems in which two new σ bonds are formed, joining the two reagents at each end. The mechanism has one step with no intermediates, and all the arrows start on π bonds and go round in a ring.



- The cycloadditions are suprafacial—they occur on one face only of each π system—and for a thermally allowed reaction there should be $4n + 2$ electrons in the mechanism, but $4n$ in a photochemical cycloaddition. These rules are dictated by orbital symmetry.
- Cycloaddition equilibria generally lie over on the right-hand side in a thermal reaction because C—C σ bonds are stronger than C—C π bonds. In a photochemical cycloaddition the product loses its π bonds and therefore its means of absorbing energy. It is therefore the kinetic product of the reaction even if it has a strained four-membered ring.

- The stereochemistry of each component is faithfully reproduced in the product—the reactions are stereospecific—and the relationship between their stereochemistries may be governed by orbital overlap to give an *endo* product.

In the next chapter we meet two more classes of pericyclic reactions: electrocyclic reactions and sigmatropic rearrangements.

Further reading

For explanations of pericyclic reactions and other reactions, using the full molecular orbital treatment, consult: Ian Fleming, *Molecular Orbitals and Organic Chemical Reactions, Student Edition*, Wiley, Chichester 2009. There is also a more comprehensive edition intended for practicing chemists, called the *Library Edition*. He has also written an Oxford Primer: *Pericyclic Reactions*, OUP, Oxford, 1999.

For a comprehensive treatment of cycloadditions in the synthesis of nitrogen heterocycles, see P. Wyatt and S. Warren, *Organic Synthesis: Strategy and Control*, Wiley, Chichester, 2007, chapter 34.

The biotin synthesis on p. 904 is described by P. Confalone and his group, *J. Am. Chem. Soc.*, 1980, **102**, 1954.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>